CS479: Machine Learning for 3D Data

3D Representations

LECTURE 2

MINHYUK SUNG

Spring 2025

KAIST

Previously in CS479

Applications: 3D Reconstruction → decoding net.

Google Maps Immersive View
https://cdn.mos.cms.futurecdn.net/P7HseGaXpSTQM2uAfSbh5Y.jpg

Previously in CS479

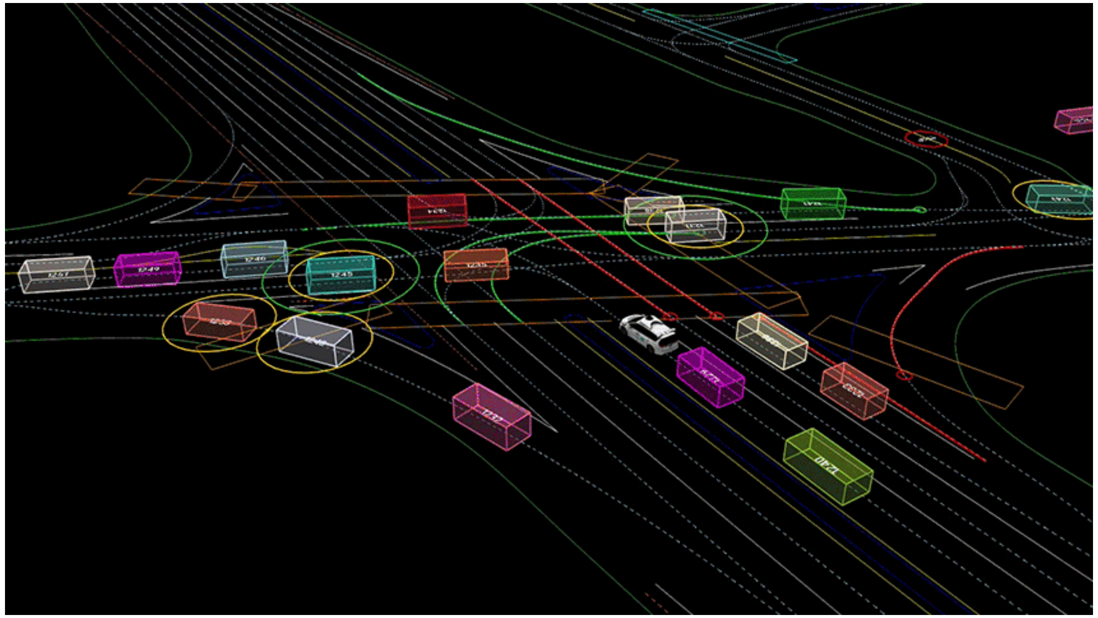
Applications: 3D Generation → decoding Net.

Wang et al., ProlificDreamer: High-Fidelity and Diverse Text-to-3D Generation with Variational Score Distillation, arXiv 2023.

Roblox

Previously in CS479

Applications: 3D Perception → encoding Net.

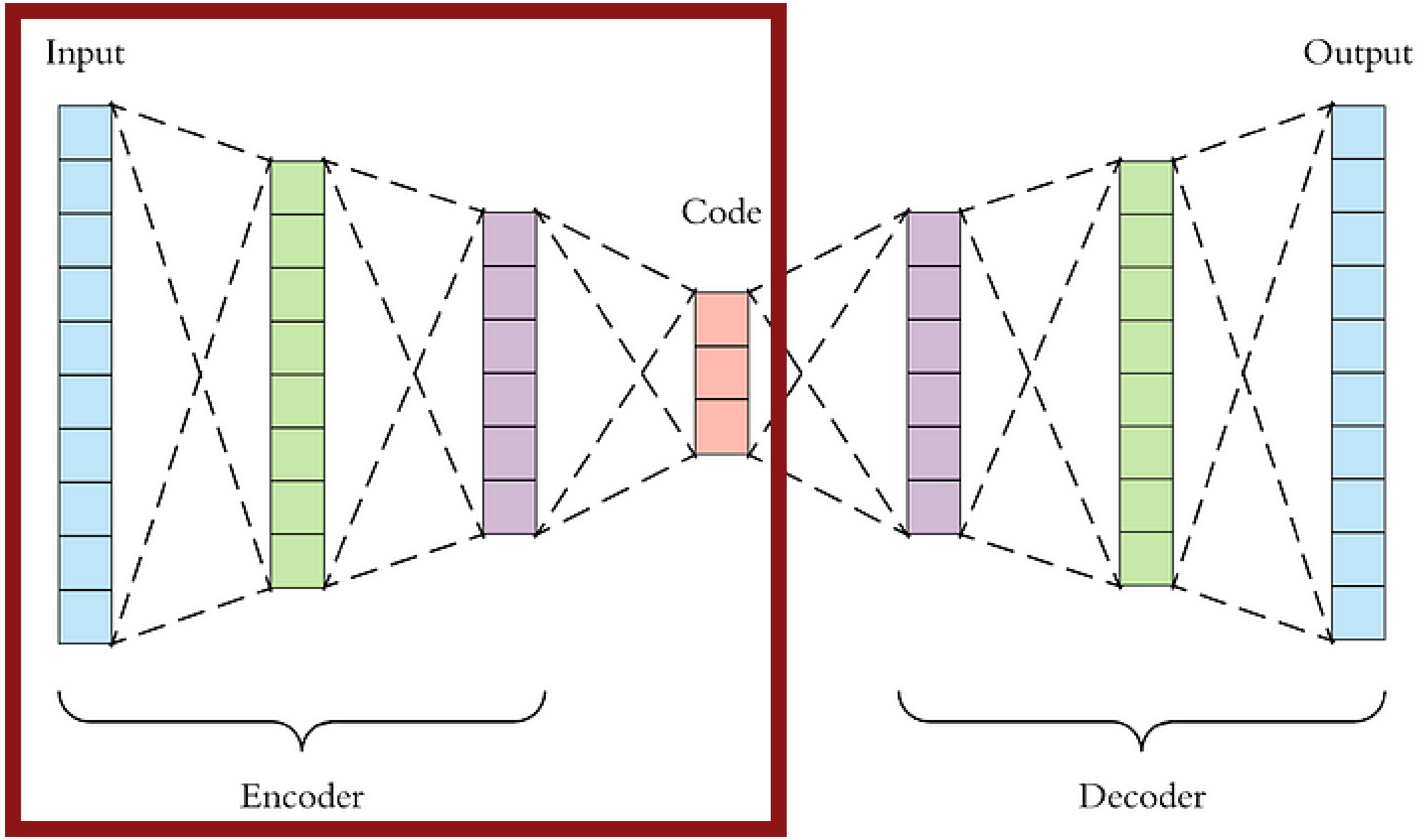


Waymo Open Dataset
https://waymo.com/blog/2021/03/expanding-waymo-open-dataset-with-interactive-scenario-data-and-new-challenges.html

AI Habitat
https://aihabitat.org/

3D Encoder

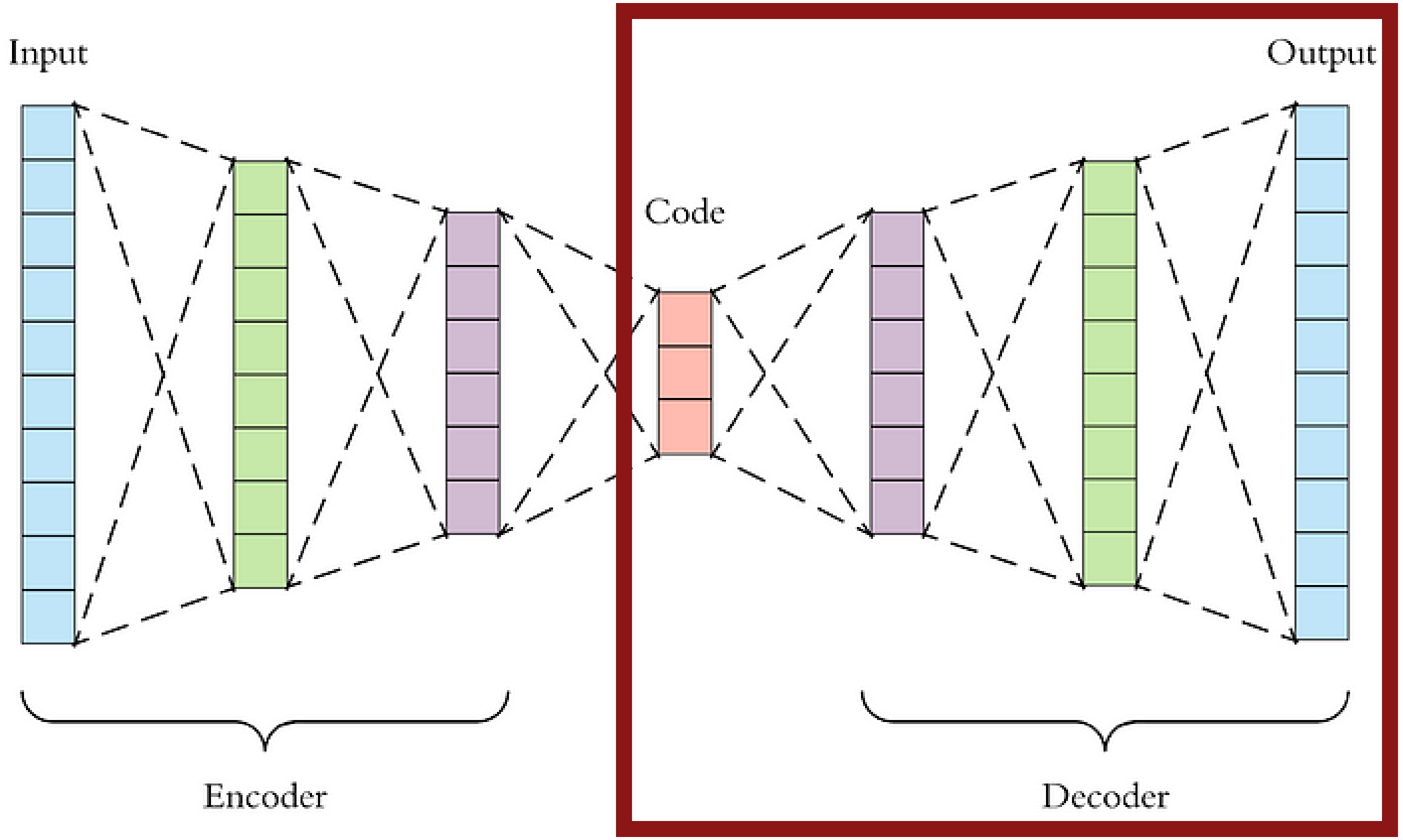
A neural network taking 3D data as input.



https://towardsdatascience.com/applied-deep-learning-part-3-autoencoders-1c083af4d798

3D Decoder

A neural network generating 3D data as output.



https://towardsdatascience.com/applied-deep-learning-part-3-autoencoders-1c083af4d798

Course Road Map

Representations

- Point clouds
- Implicit representation
- Multi-view images to 3D
- Hybrid representations
- Meshes
- ~~CAD~~
- Representation Conversion

Applications

- 3D perception
- Reconstruction
- Manipulation
- ~~Generation~~

3D Representations

Texts

Text

"The cat sat on the mat."

↓

Tokens

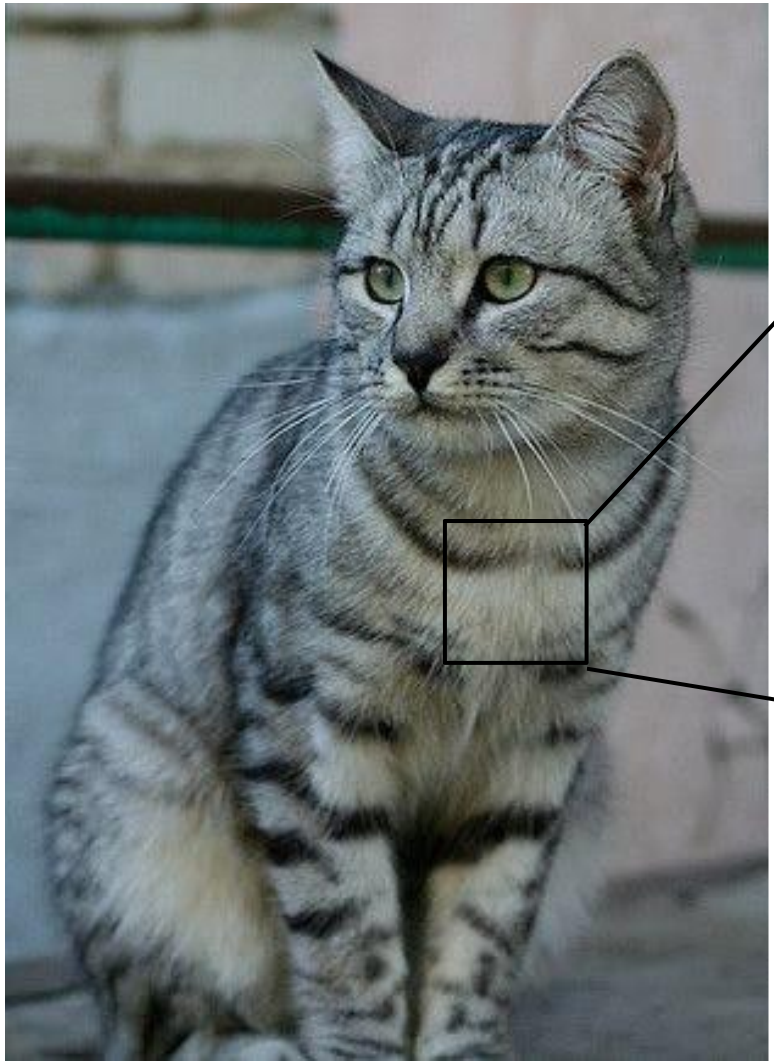
"the", "cat", "sat", "on", "the", "mat", "."

↓

Vector encoding of the tokens

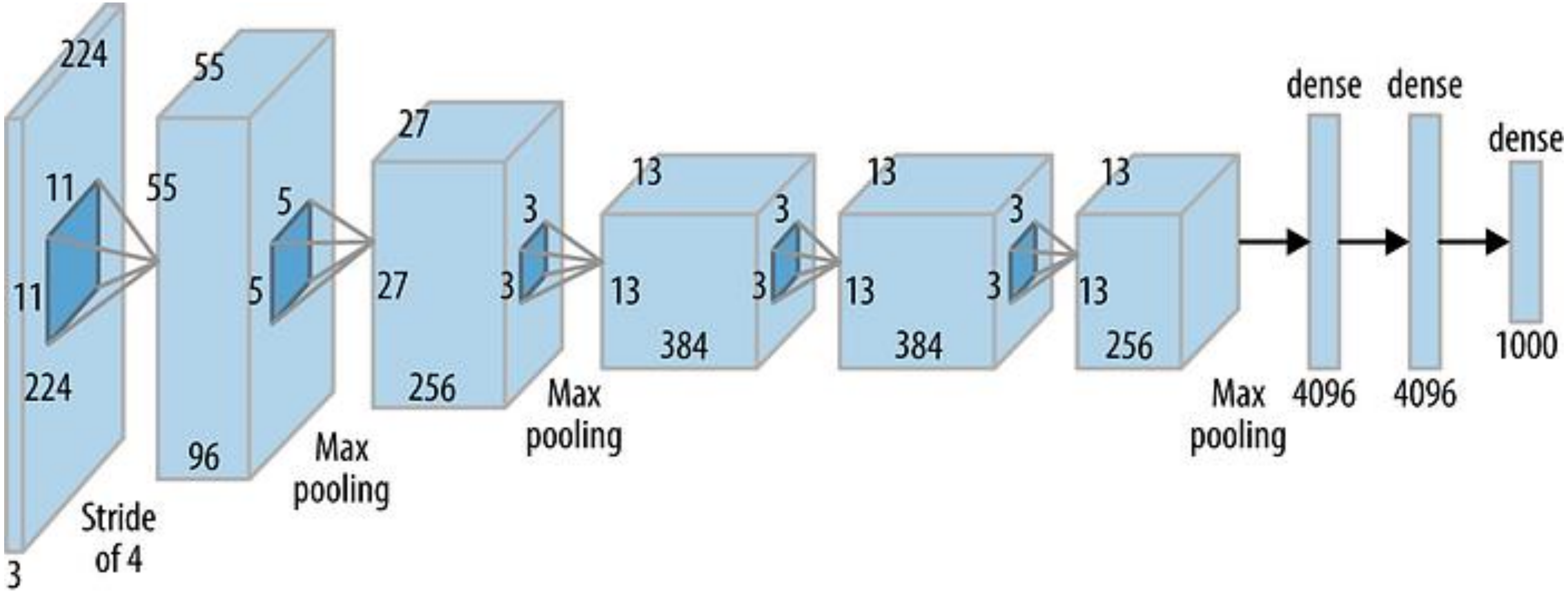
0.0	0.0	0.4	0.0	0.0	1.0	0.0
0.5	1.0	0.5	0.2	0.5	0.5	0.0
1.0	0.2	1.0	1.0	1.0	0.0	0.0
the	cat	sat	on	the	mat	.

Images

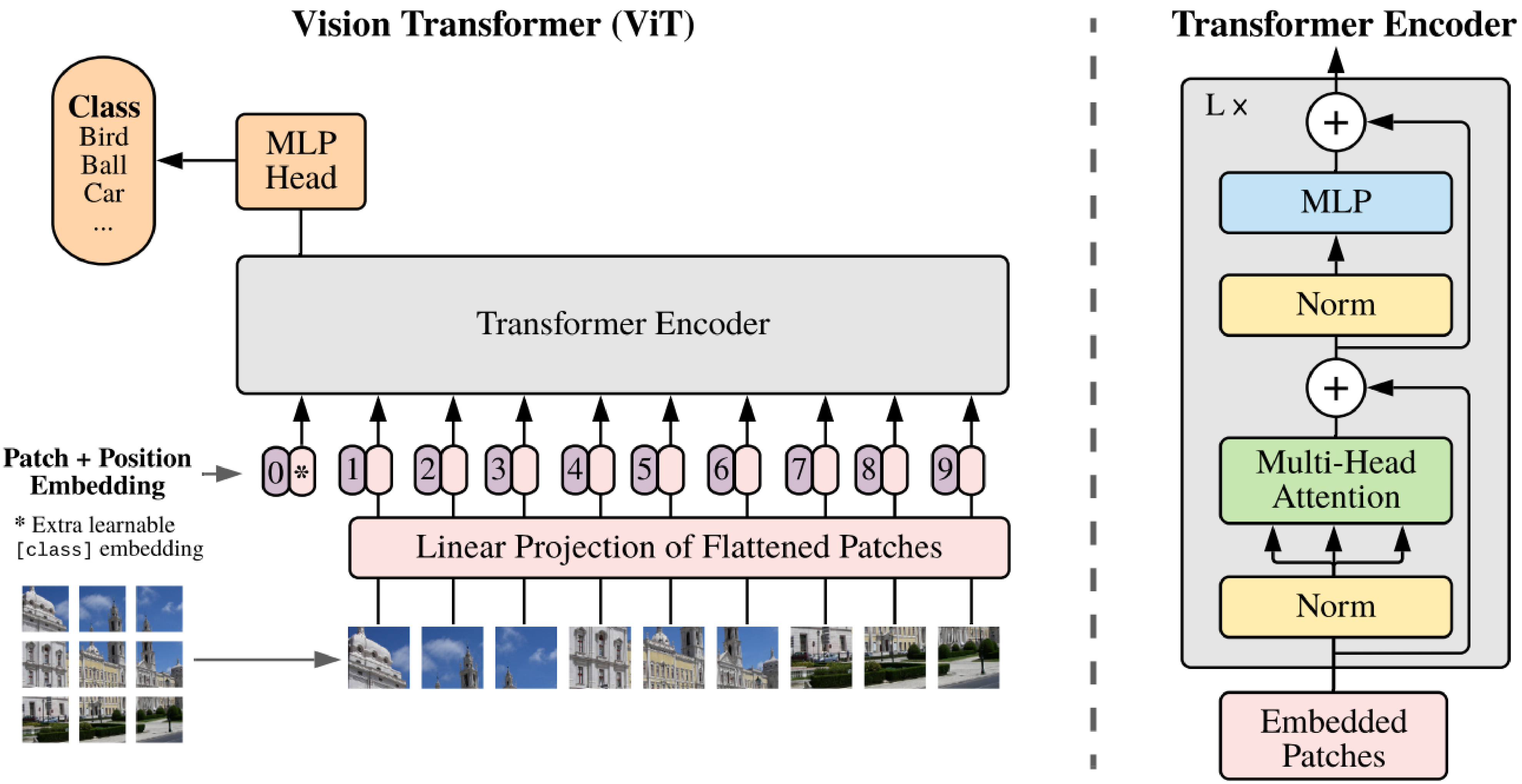


```
[[105 112 108 111 104  99 106  99  96 103 112 119 104  97  93  87]
 [ 91  98 102 106 104  79  98 103  99 105 123 136 110 105  94  85]
 [ 76  85  90 105 128 105  87  96  95  99 115 112 106 103  99  85]
 [ 99  81  81  93 120 131 127 100  95  98 102  99  96  93 101  94]
 [106  91  61  64  69  91  88  85 101 107 109  98  75  84  96  95]
 [114 108  85  55  55  69  64  54  64  87 112 129  98  74  84  91]
 [133 137 147 103  65  81  80  65  52  54  74  84 102  93  85  82]
 [128 137 144 140 109  95  86  70  62  65  63  63  60  73  86 101]
 [125 133 148 137 119 121 117  94  65  79  80  65  54  64  72  98]
 [127 125 131 147 133 127 126 131 111  96  89  75  61  64  72  84]
 [115 114 109 123 150 148 131 118 113 109 100  92  74  65  72  78]
 [ 89  93  90  97 108 147 131 118 113 114 113 109 106  95  77  80]
 [ 63  77  86  81  77  79 102 123 117 115 117 125 125 130 115  87]
 [ 62  65  82  89  78  71  80 101 124 126 119 101 107 114 131 119]
 [ 63  65  75  88  89  71  62  81 120 138 135 105  81  98 110 118]
 [ 87  65  71  87 106  95  69  45  76 130 126 107  92  94 105 112]
 [118  97  82  86 117 123 116  66  41  51  95  93  89  95 102 107]
 [164 146 112  80  82 120 124 104  76  48  45  66  88 101 102 109]
 [157 170 157 120  93  86 114 132 112  97  69  55  70  82  99  94]
 [130 128 134 161 139 100 109 118 121 134 114  87  65  53  69  86]
 [128 112  96 117 150 144 120 115 104 107 102  93  87  81  72  79]
 [123 107  96  86  83 112 153 149 122 109 104  75  80 107 112  99]
 [122 121 102  80  82  86  94 117 145 148 153 102  58  78  92 107]
 [122 164 148 103  71  56  78  83  93 103 119 139 102  61  69  84]]
```

Convolutional Neural Network



Transformers

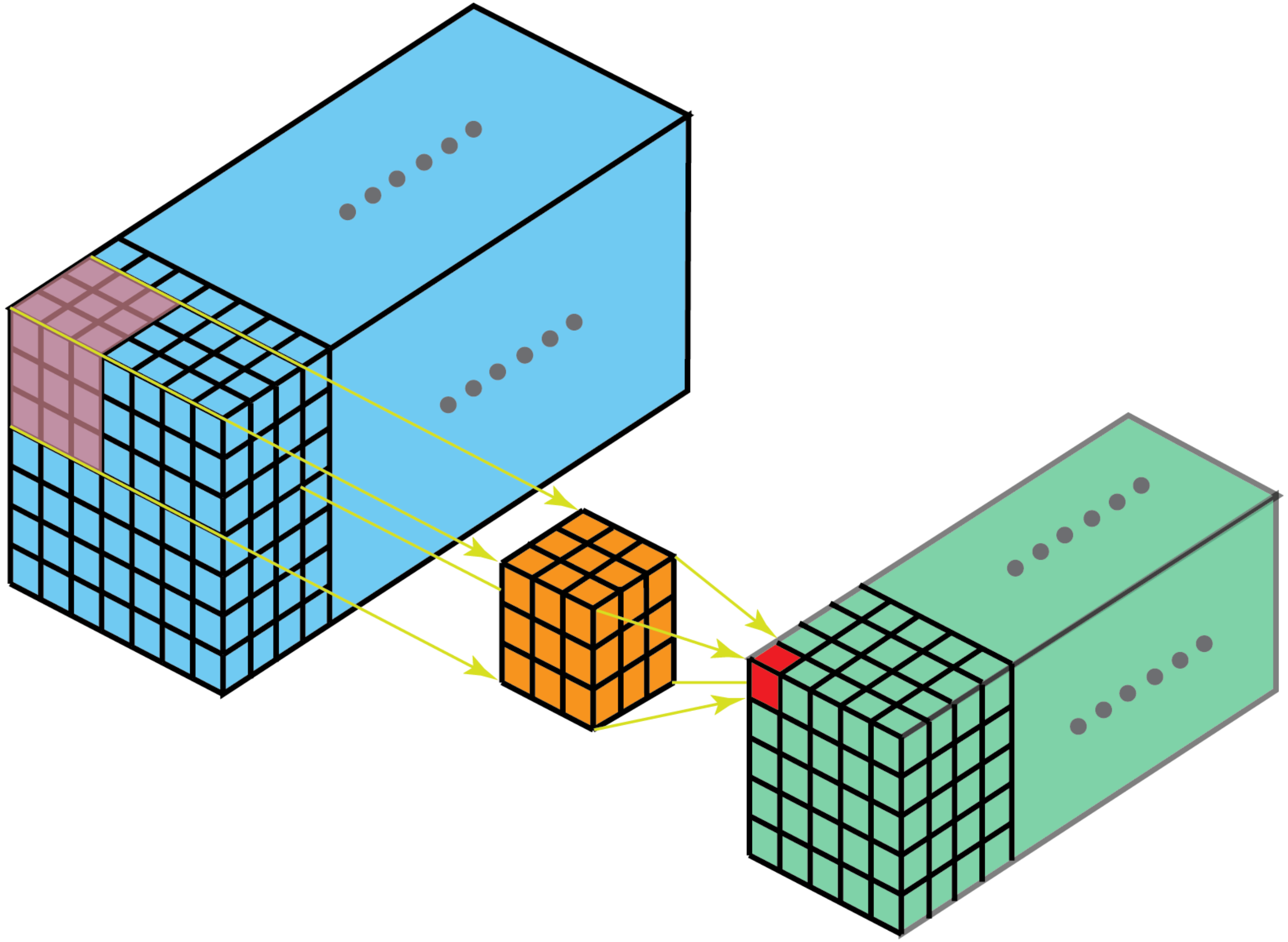


3D Grid (Voxels)

Voxel : Volume(부피) + Element

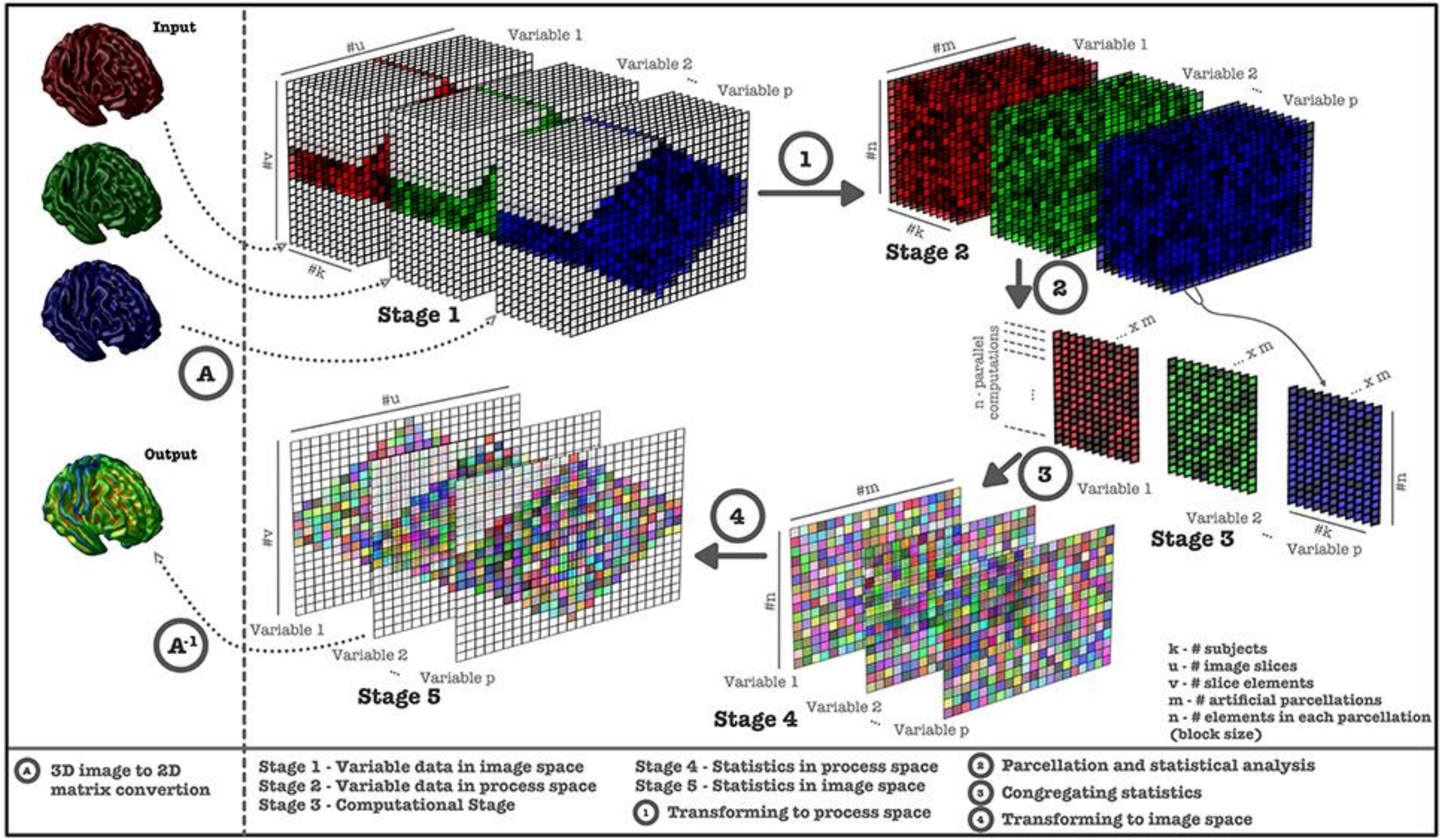
: 3차원 공간의 Pixel.

3D Grid



https://towardsdatascience.com/a-comprehensive-introduction-to-different-types-of-convolutions-in-deep-learning-669281e58215

Medical Imaging



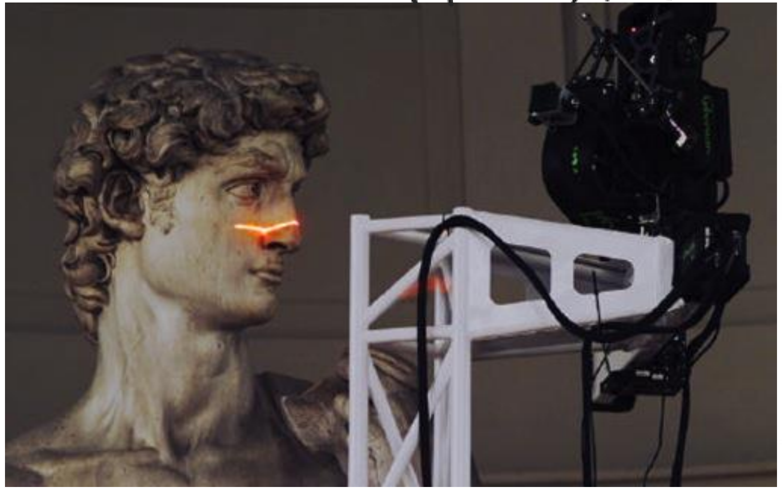
Mathotaarachchi et al., VoxelStats, 2016.

We’re interested in 2D surfaces in 3D space.

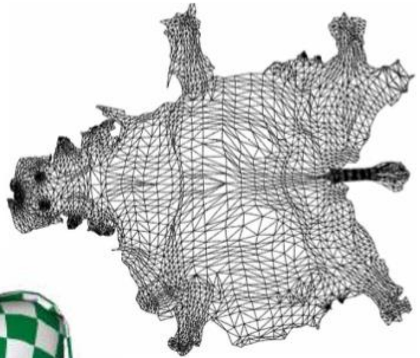
ㅇ 3d info from 3d scan. or 3d model. → issue: volume info에 관심 X. surface info에 관심 O.

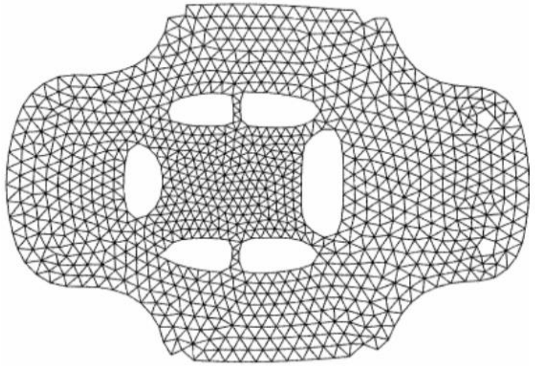
ㅇ we don't know and care about inside info.

→ scene에서도 비슷. (3d space 대부분은 empty!)



The Digital Michelangelo Project





Liu et al., A Local/Global Approach to Mesh Parameterization, SGP 2008.

How we encode all 3d Info into the voxels

→ one way is encoding occupancy Info like which voxel is inside or which voxel is outside.

Encoding Surface Info

→ checking what are the voxels where the surface is basically passing.

Encode voxel having the surface is 1.

other voxels are 0. (empty voxel 이니까).

2d vs 3d.

2d : 해상도 2배 up → 전체 픽셀 수 2(가로) X 2(세로) = 4배 up.

3d : // → 전체 Voxel 수 2X2X2 = 8배 up.

☆ 전체 복셀 개수와 표면 구성 복셀 개수가 늘어나는 속도가 다름.

공간 부피에 비례. $O(N^3)$ 물체 겉넓이에 비례. $O(N^2)$

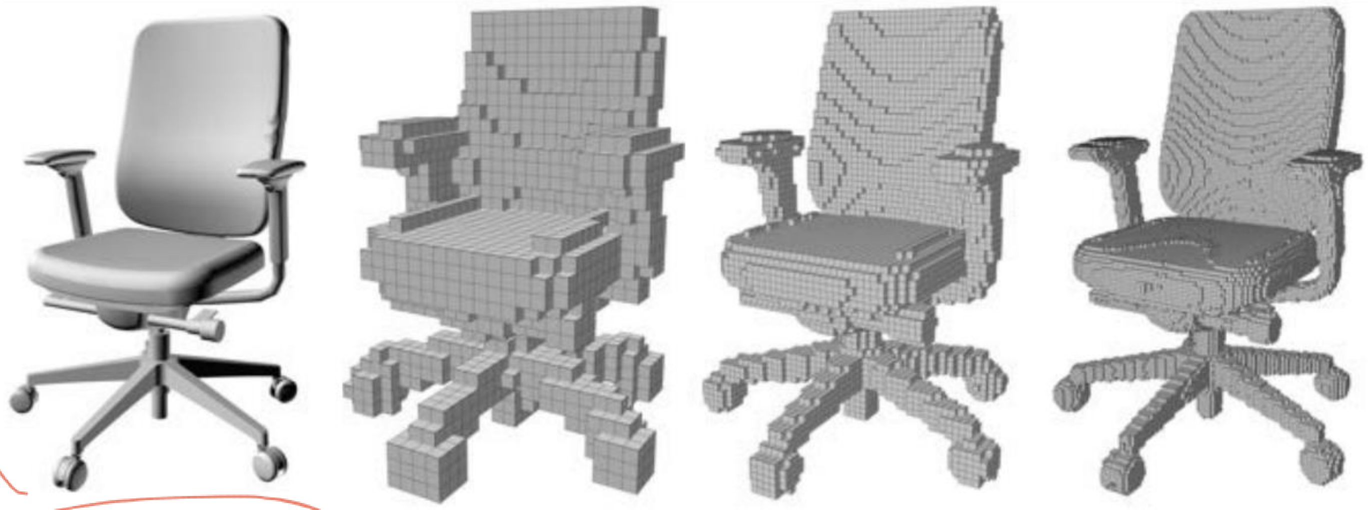
$$\frac{\text{Surface Voxels}}{\text{Total voxels}} \approx \frac{N^2}{N^3} = \frac{1}{N}.$$

해상도(N)가 커질수록 전체 공간에서 표면 Voxel이 차지하는 비율은 1/N에 비례해서 계속 작아짐.

3D Convolution

"고해상도로 갈수록 관심있는 '표면' 데이터는 전체 공간의 아주 극소수에만 존재. 99%는 내부/외부의 empty space 나타내는 쓸모없는 voxel."

- Each voxel contains a binary value (indicating whether the voxel is on the surface or not), and also, most voxels are empty. → whole representation becomes very inefficient and redundant.
- Huge waste of computation.



Occupied voxel ratio	10.41%	5.09%	2.41%
Resolution	32	64	128

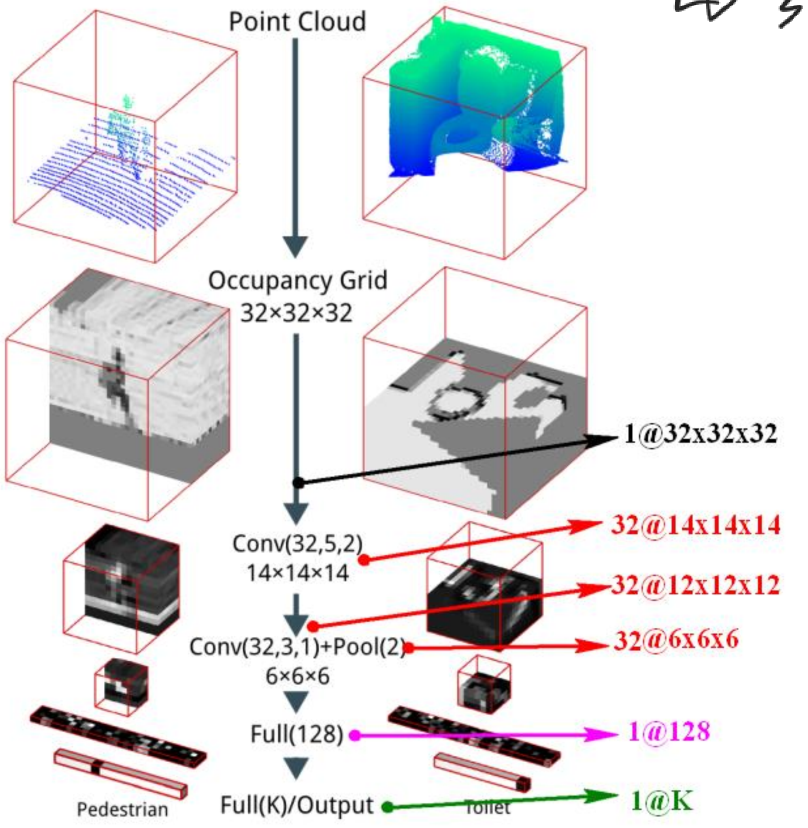
실제 표면정보를 가지고 있는 voxel

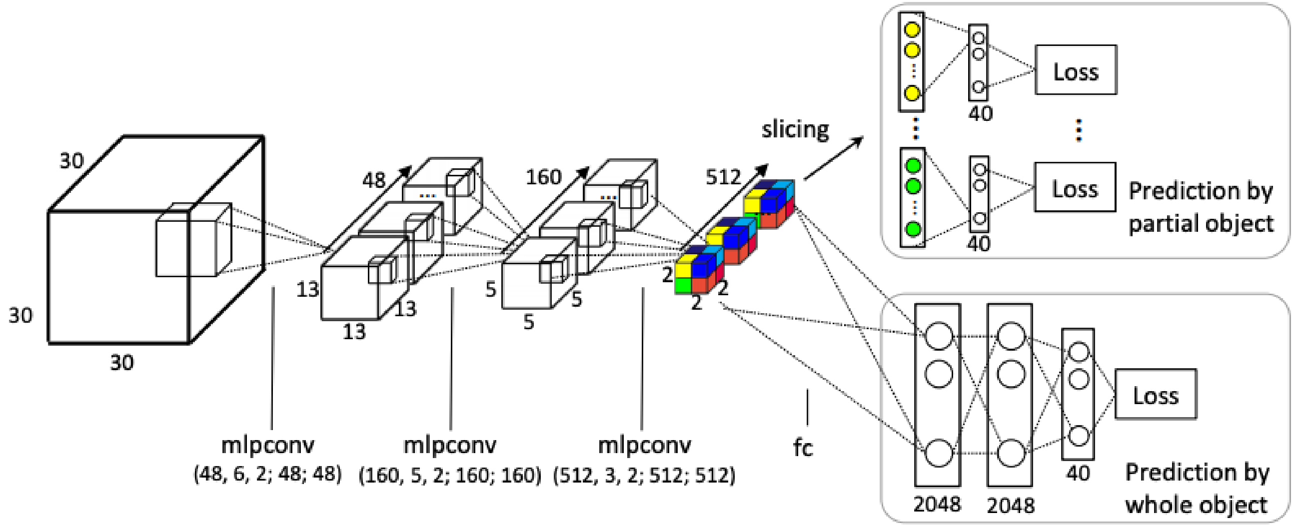
해상도 올릴수록 표면에 있는 voxel 수가 sparse 해짐.
(sparse representation)

3D CNNs

(–) Takes a huge amount of memory and time in training.

↳ 3d volume은 기본적으로 very sparse Information을 가지기 때문.

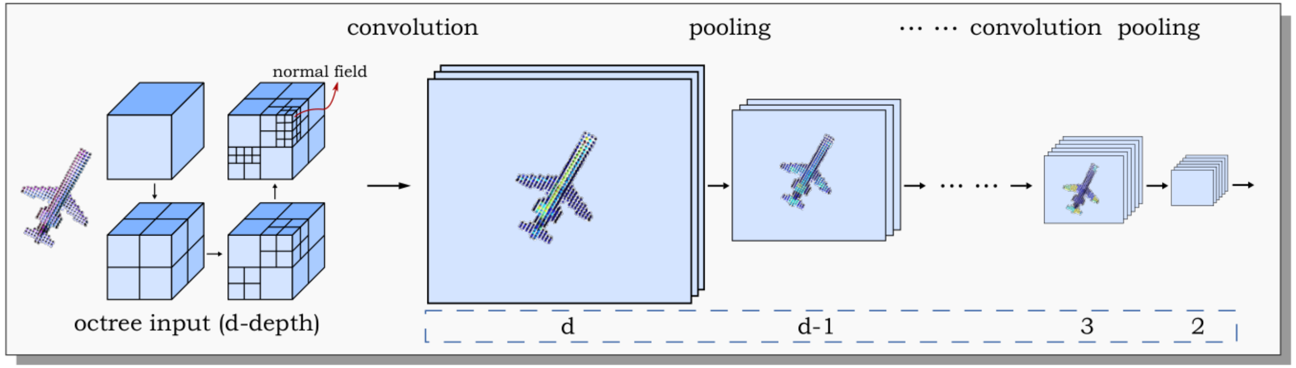




3D CNNs

Bigger voxels in the empty space.

Architectures using adaptive data structure



Wang et al., O-CNN, SIGGRAPH 2017.

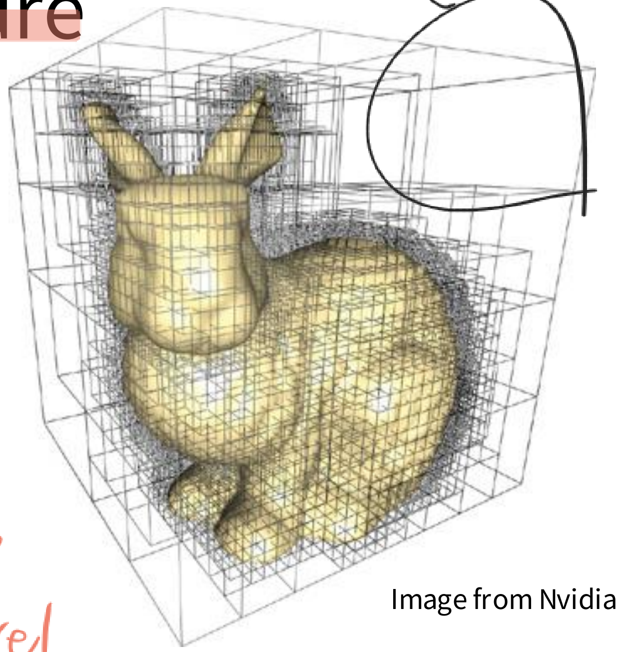
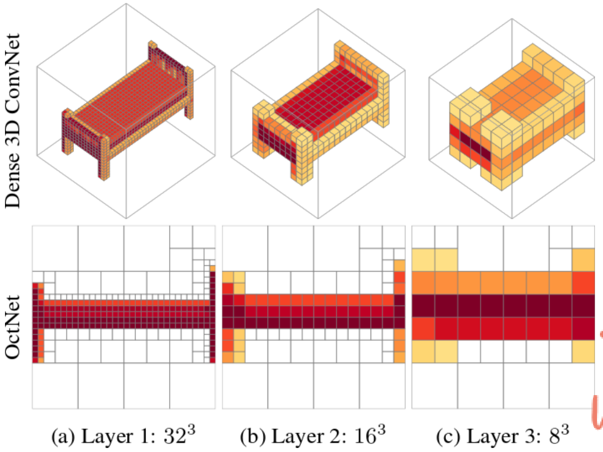


Image from Nvidia

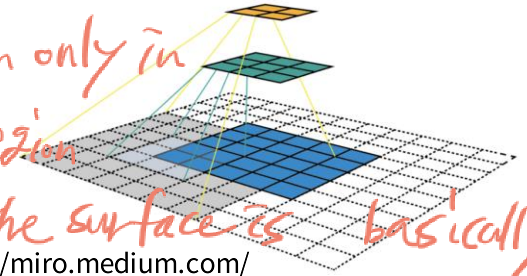
Convolution 연산에서 adaptive resolution 을 사용해야 해서 구현은 더 어려워짐. 하지만 doable

a No surface info
→ bigger voxel.

a Increasing resolution only in some kind of local region where we know that the surface is basically located.



Riegler et al., O-CNN, CVPR 2017.



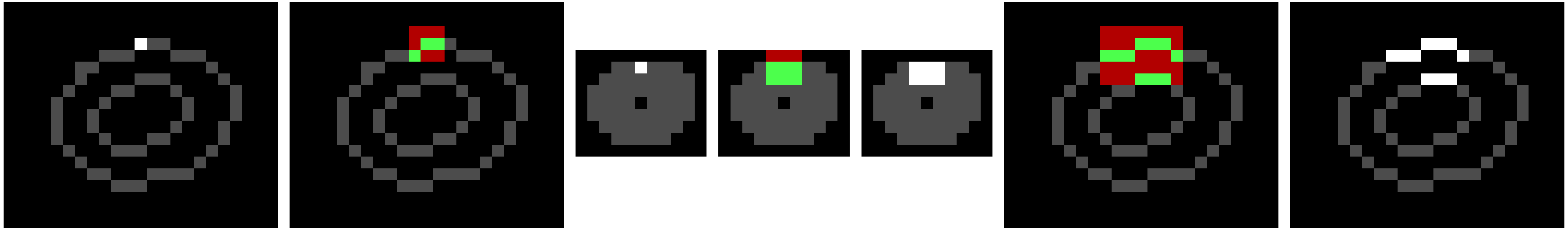
https://miro.medium.com/

Grahamet al., 3D Semantic Segmentation with Submanifold Sparse Convolutional Networks, CVPR 2018.

SparseConvNet [Grahamet al., 2018]

- Compute convolutions only in the active areas.
- Still takes lots of memory and time in training.

surface가 located 된.
region만.

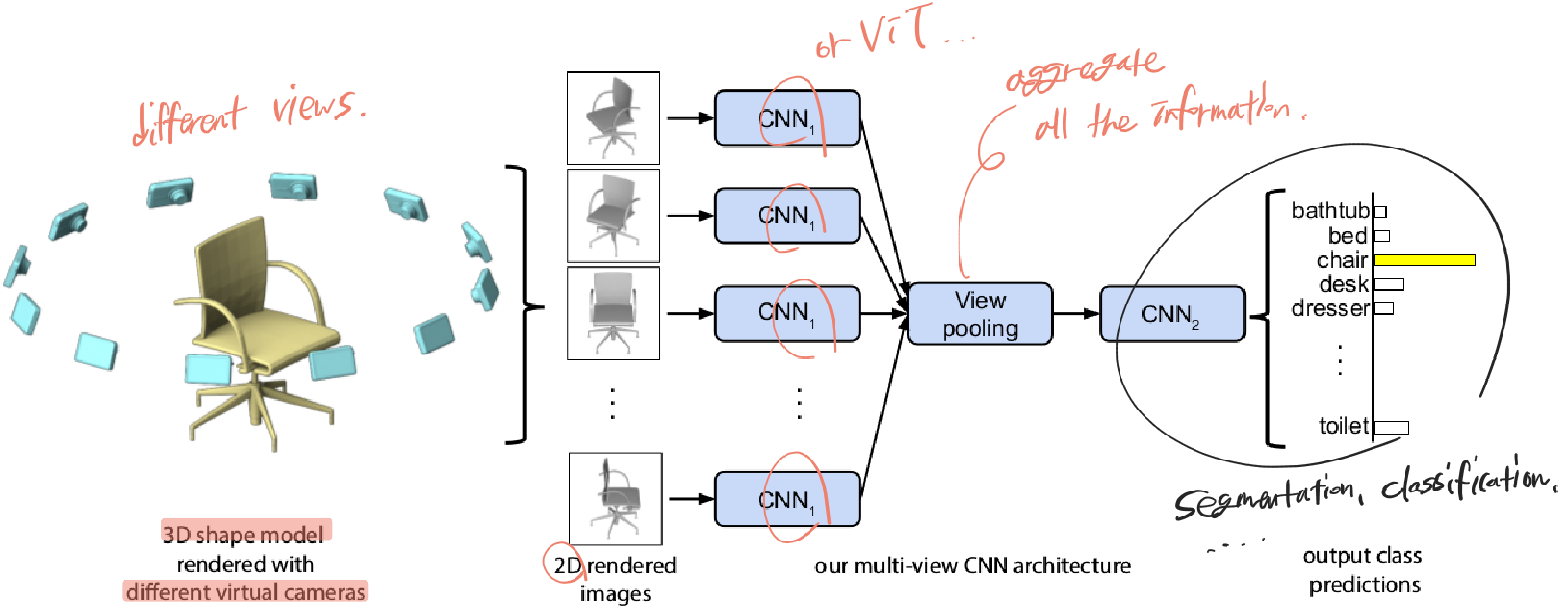


Multi-View Images

2015부터는 모든 3d data를 이미지 set으로 처리하는 방식 등장~.

↳ Multi-view Image representation 시도한 것.

Multi-View CNN [Su et al., 2015]



3D Object Classification

3d Conv

2d Conv.

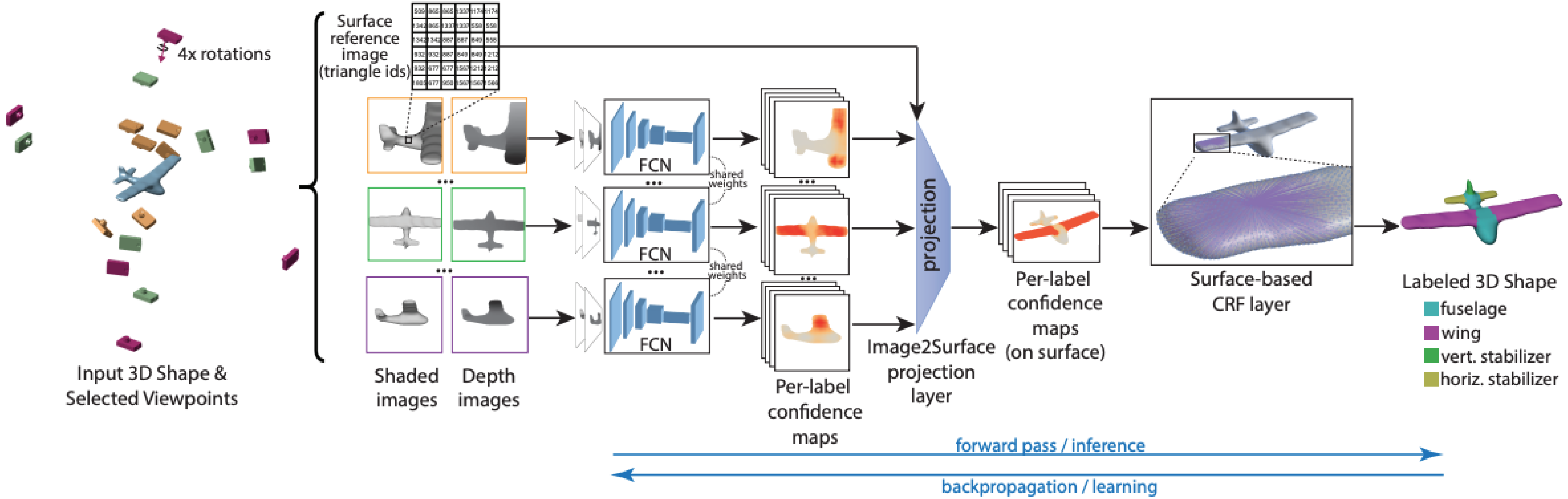
	input	#views	accuracy avg. class	accuracy overall
SPH [12]	mesh	-	68.2	-
3DShapeNets [29]	volume	1	77.3	84.7
VoxNet [18]	volume	12	83.0	85.9
Subvolume [19]	volume	20	86.0	**89.2**
LFD [29]	image	10	75.5	-
MVCNN [24]	image	80	**90.1**	-

3d data 처리를 위한 special한 Net이 필요없음!!

multi-view.

Dataset: ModelNet40
Metric: 40-class classification accuracy (%)

3D Segmentation with 2D Neural Networks



Multi-View Images

- (+) Especially good for processing **appearance** information like color, texture, and material.

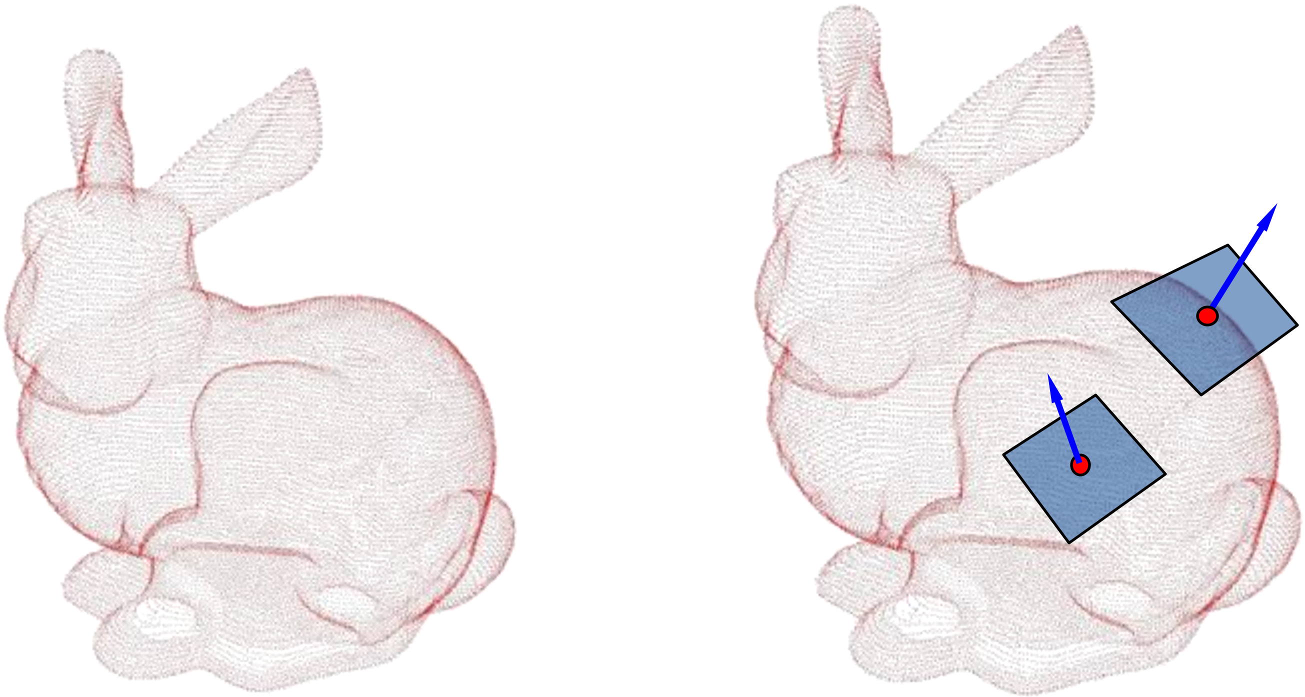
- (–) Requires lots of images for high accuracy and thus takes lots of memory and time.

- (–) May not be able to capture geometric details.

Point Cloud

Point Cloud

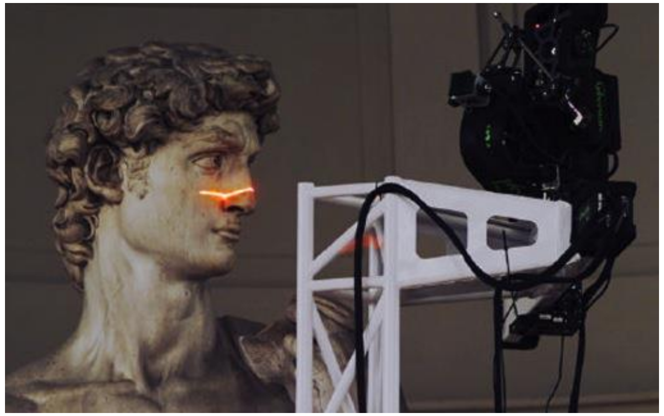
- The simplest representation: **only points**, no connectivity.
- Collection of (x,y,z) coordinates, possibly with normal.

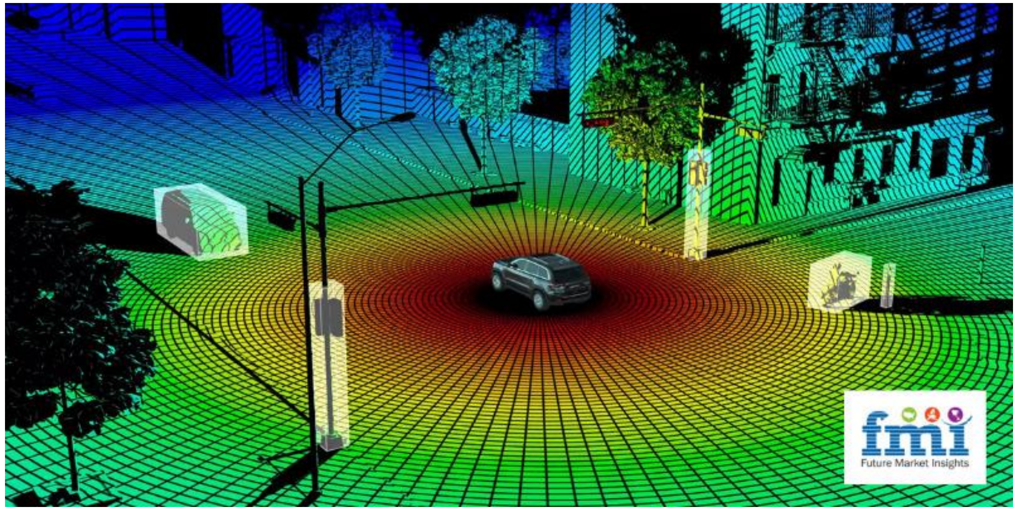


Stanford bunny

Point Cloud

- Nearly all 3D scanning devices produce point clouds.





https://techbullion.com/wp-content/uploads/2021/12/Mobile-LiDAR-Scanner.jpg

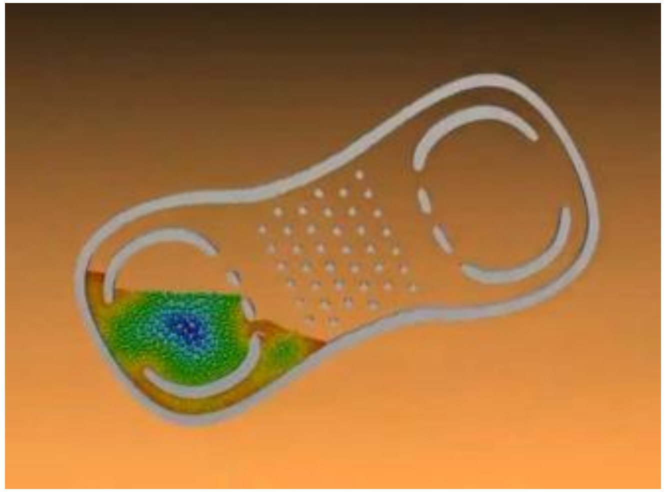
Point Cloud

- Nearly all 3D scanning devices produce point clouds.
- Sometimes, easier to handle.

Fracturing Solids

Meshless Animation of Fracturing Solids
Pauly et al., SIGGRAPH ‘05

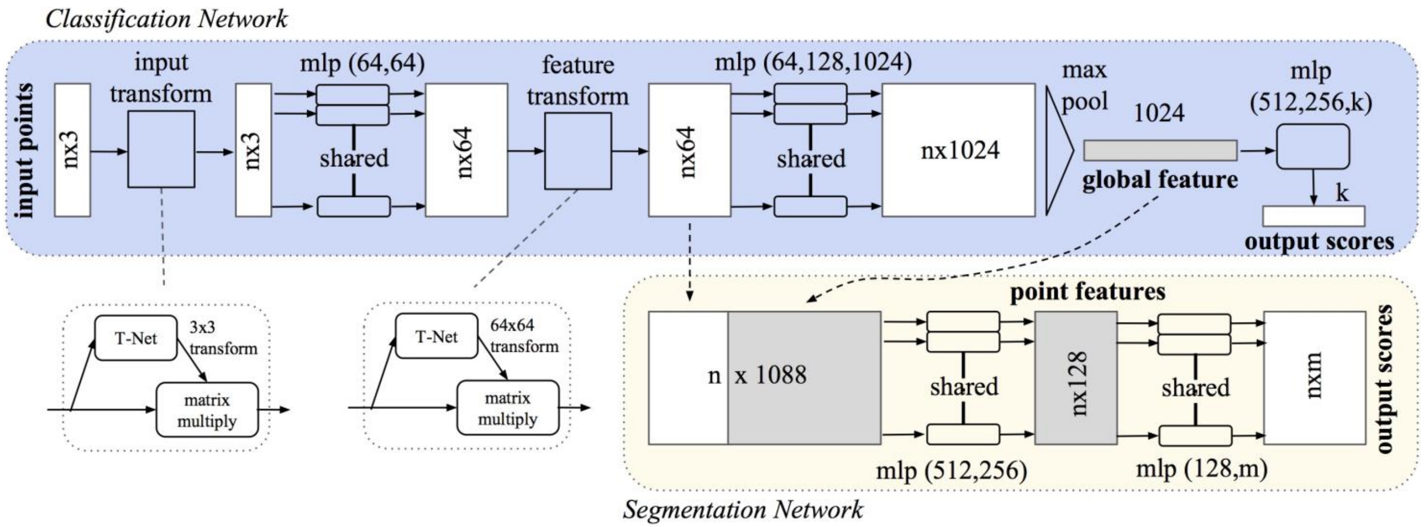
Fluids



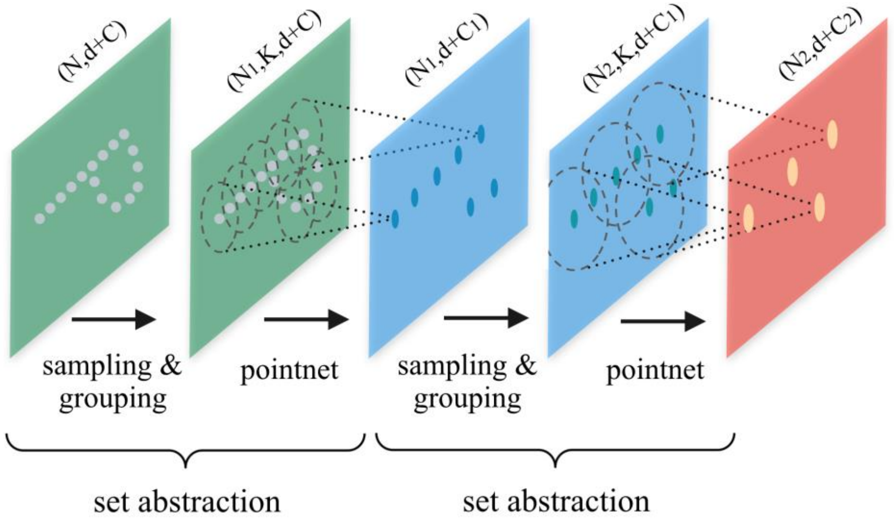
Adaptively sampled particle fluids,
Adams et al. SIGGRAPH ‘07

Neural Networks for Point Clouds

- (+) Fast, easy to implement (relatively).
- (+) The most popular architectures.



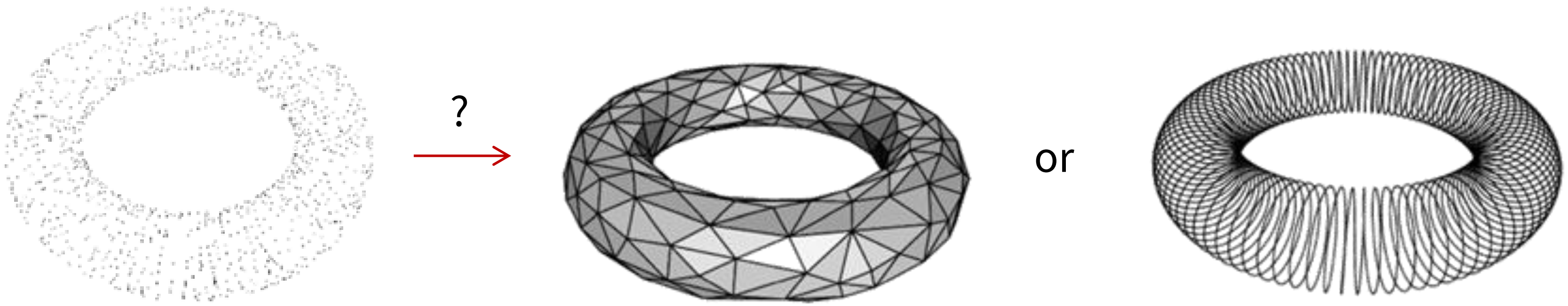
Qi et al., PointNet, CVPR 2017.



Qi et al., PointNet++, NeurIPS 2017.

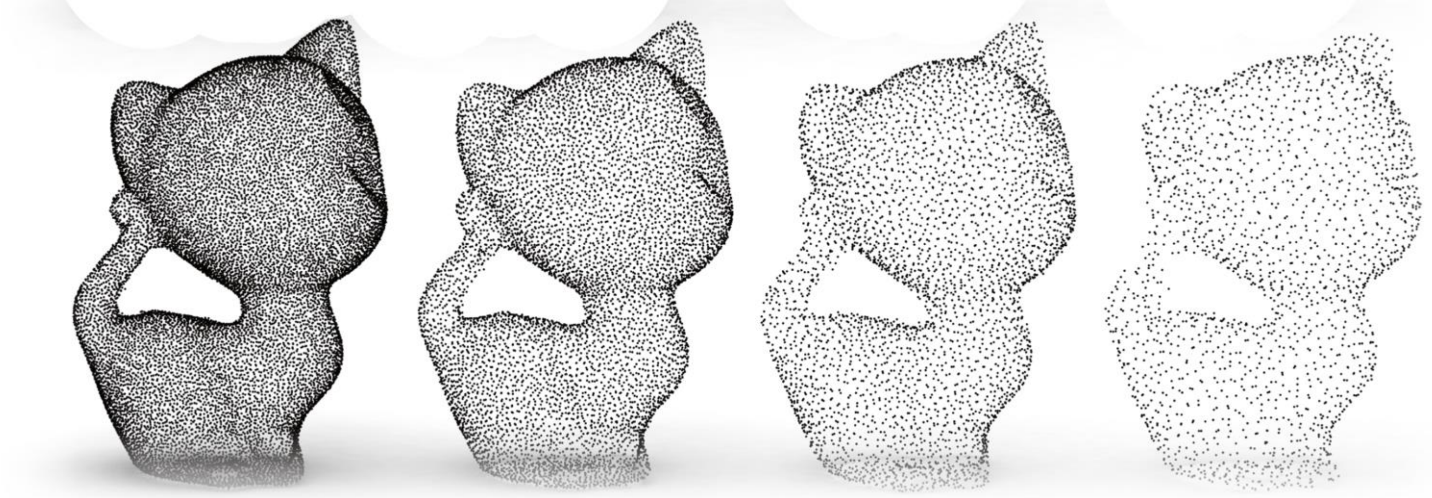
Point Cloud

- (–) No surface/topology information;
  needs to be converted to the other representation for downstream applications.



Point Cloud

- (–) No surface/topology information; needs to be converted to the other representation for downstream applications.
- (–) Weak approximation power; requires many points for the details.



Li et al., PU-GAN, ICCV 2019.

Polygon Mesh

Polygon Mesh

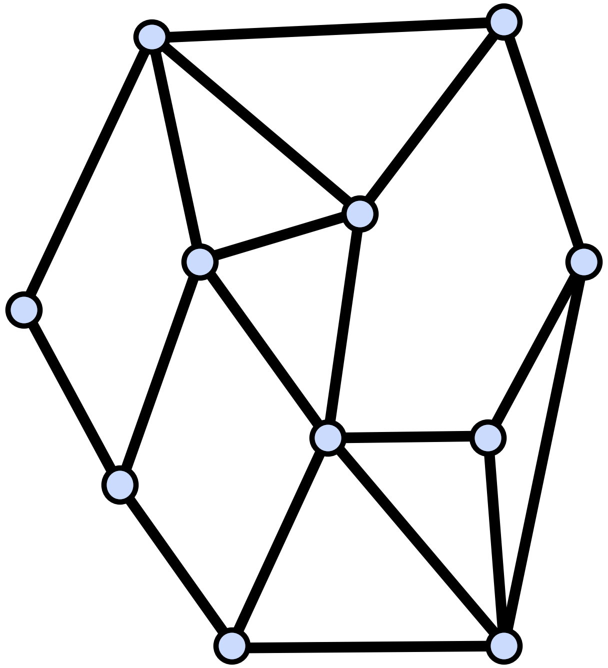
- The most popular representation for shapes in graphics.
- A compact form representing surfaces.
- A graph-like structure but not the same.

https://www.3dcadbrowser.com/3d-model/people-collection-low-poly-62652

Polygon Mesh

- A polygon mesh is a collection of vertices, edges and faces that defines the shape of a polyhedral object.

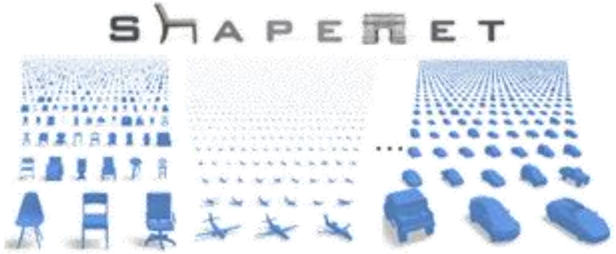
- A triangle mesh is a special case when all the faces are triangles.



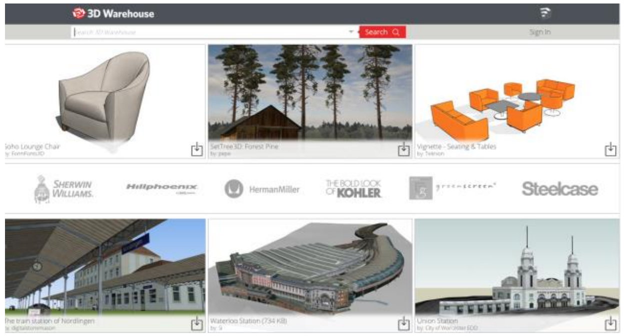
$$M = \langle V, E, F \rangle$$

vertices edges faces

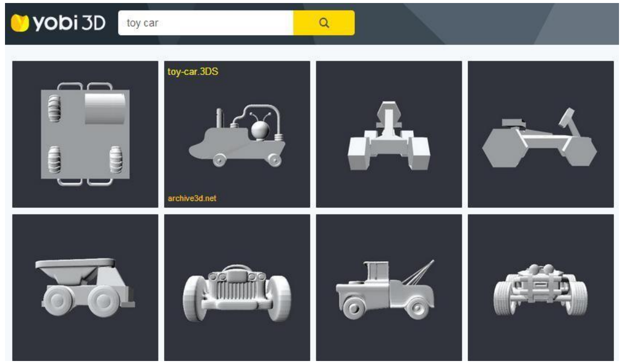
Online Repositories of 3D Meshes



ShapeNet



3D Warehouse

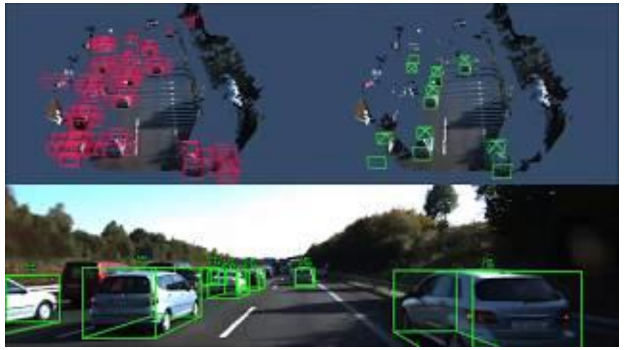


Yobi 3D

SceneNN

Redwood Dataset

ScanNet

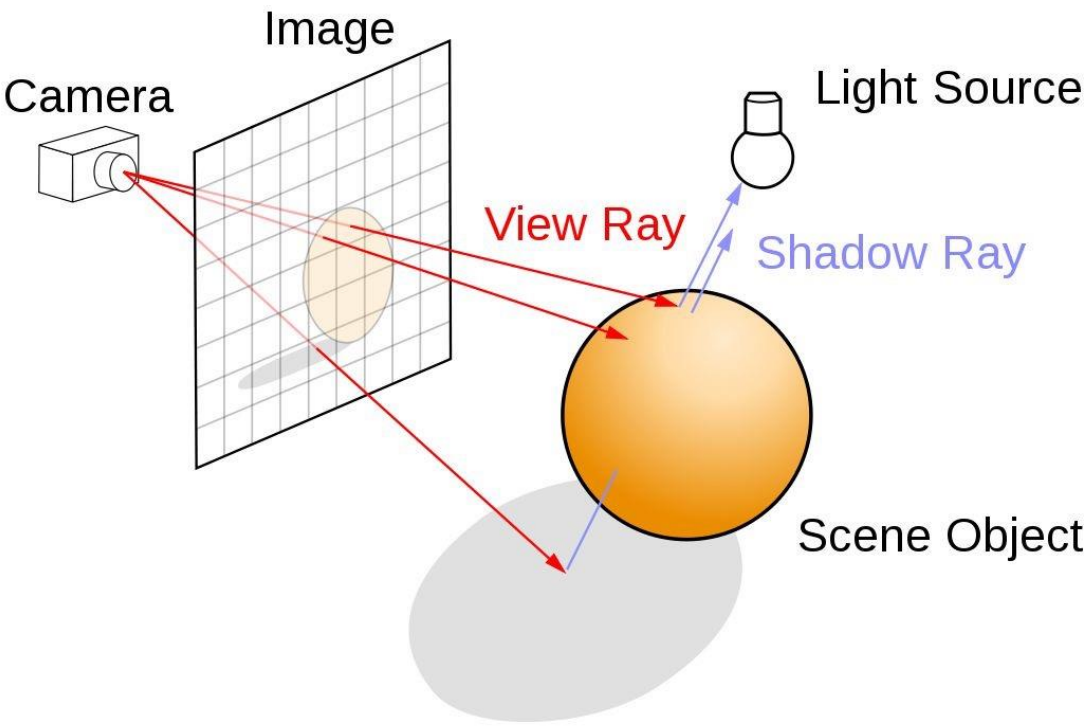


KITTI 3D

Dynamic MPI-FAUST

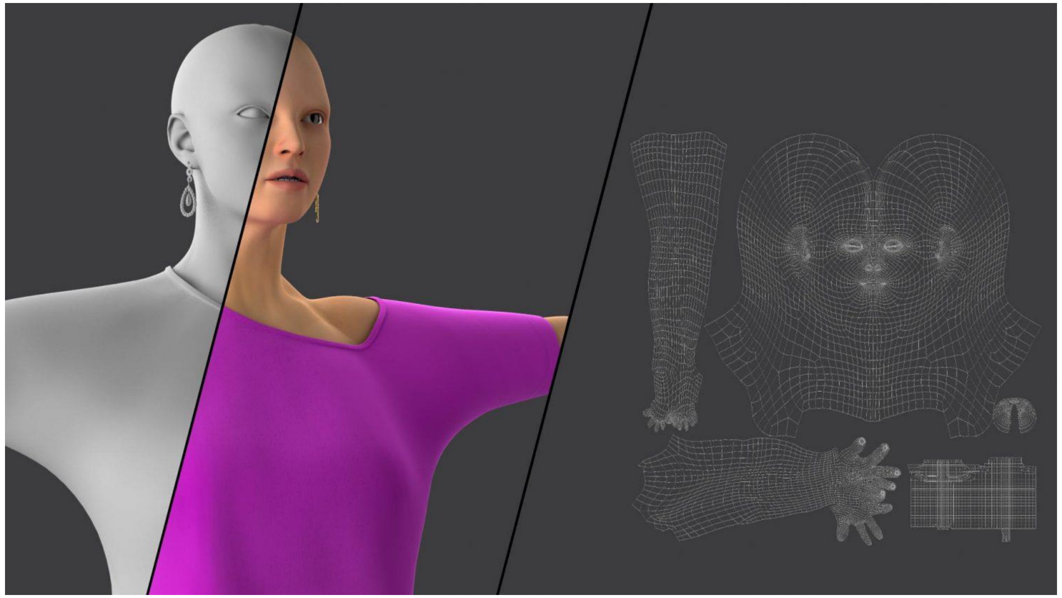
Polygon Mesh

- (+) Good for many applications:
  - Rendering

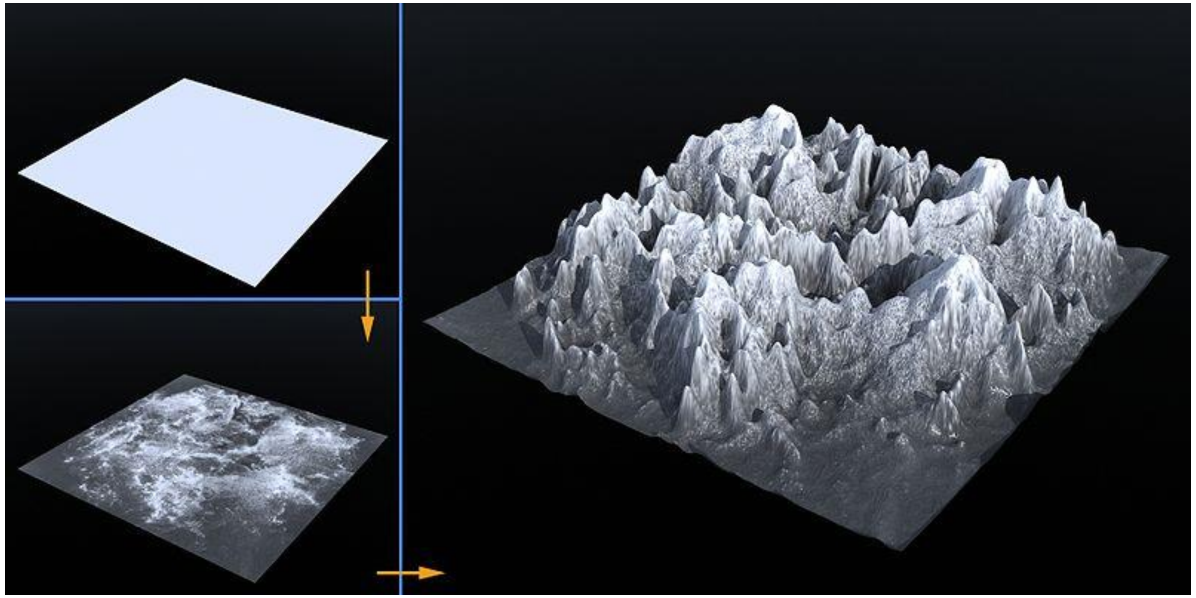


Polygon Mesh

- (+) Good for many applications:
  - Rendering
  - Texturing



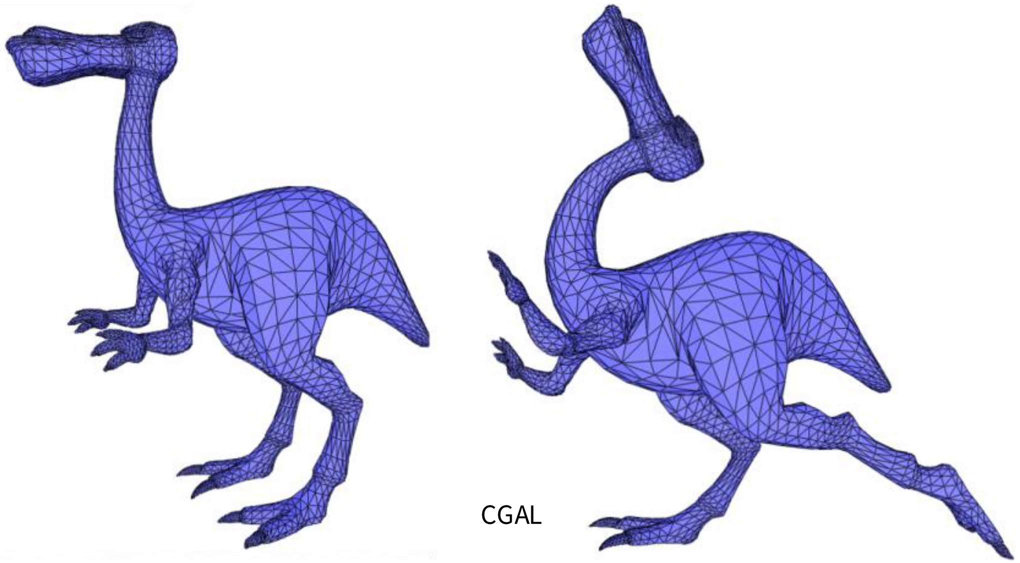
https://dreamfarmstudios.com/blog/getting-to-know-3d-texturing-in-animation-production/



https://commons.wikimedia.org/wiki/File:Displacement_Mapping.jpg

Polygon Mesh

- (+) Good for many applications:
  - Rendering
  - Texturing
  - Deformation / Manipulation

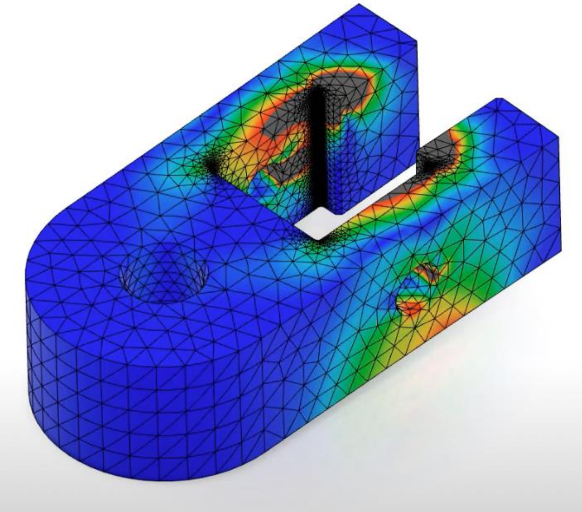


CGAL

Polygon Mesh

- (+) Good for many applications:
  - Rendering
  - Texturing
  - Deformation / Manipulation
  - Simulation

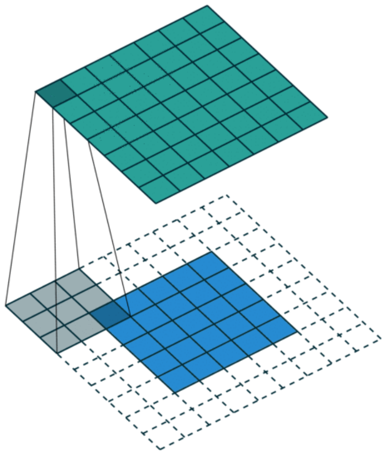
https://3dmodelsworld.com/maya-bullet-physics-simulation-tutorial-wrecking-ball-animation-active-and-passive-rigid-body/



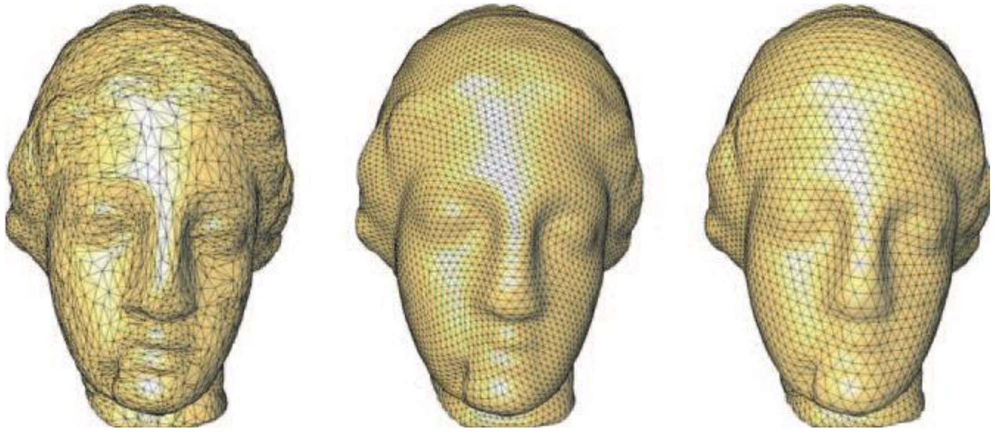
https://www.youtube.com/watch?v=HKL8mQO1iuU

Polygon Mesh

- (+) Good for many applications:
  - Rendering
  - Texturing
  - Deformation / Manipulation
  - Simulation
- (–) Irregular structure



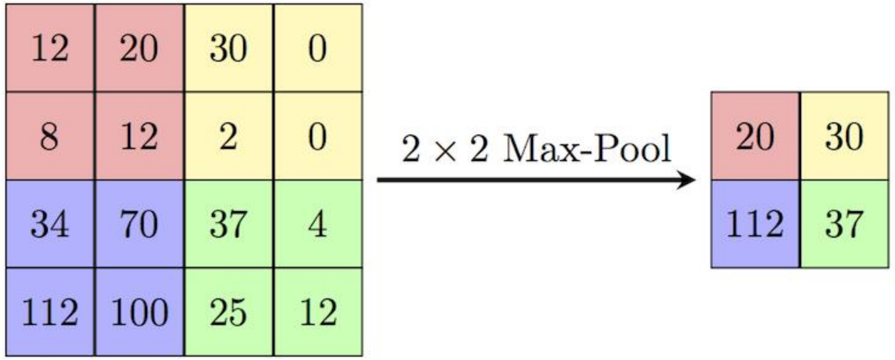
https://towardsdatascience.com/convolution-vs-correlation-af868b6b4fb5



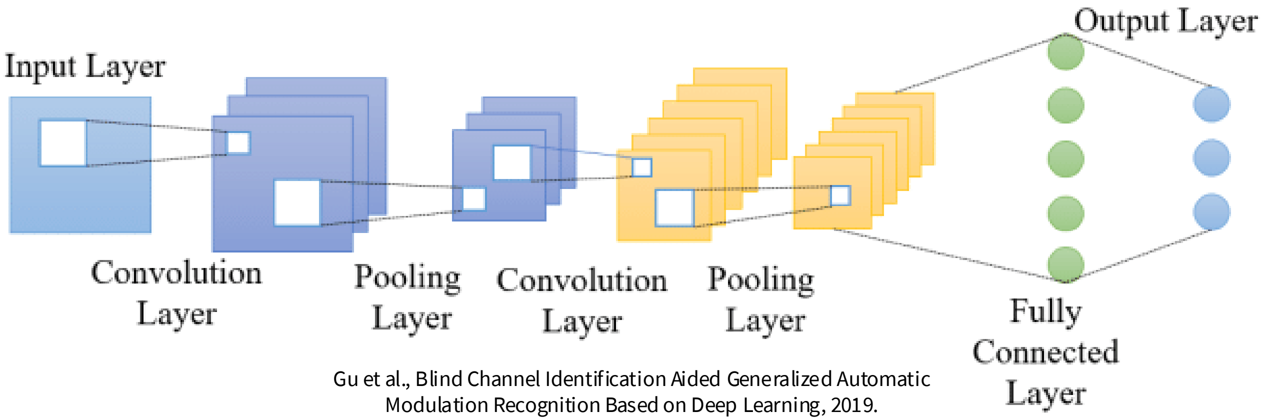
Alliez et al., Recent Advances in Remeshing of Surfaces.

Pooling in Neural Network

Aggregating information while progressively reducing the resolution of the data.



https://paperswithcode.com/method/max-pooling



Gu et al., Blind Channel Identification Aided Generalized Automatic Modulation Recognition Based on Deep Learning, 2019.

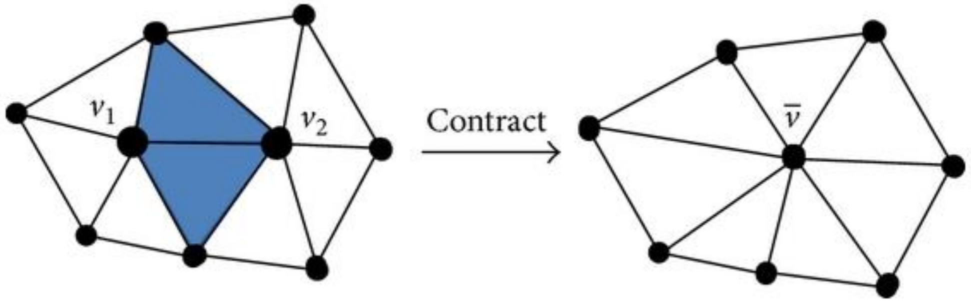
Garland and Heckbert, Surface Simplification Using Quadric Error Metrics, SIGGRAPH 1997.

Pooling Operation for Polygon Mesh

How can the resolution of a polygon mesh be decreased?

Through the process of iterative edge contraction.

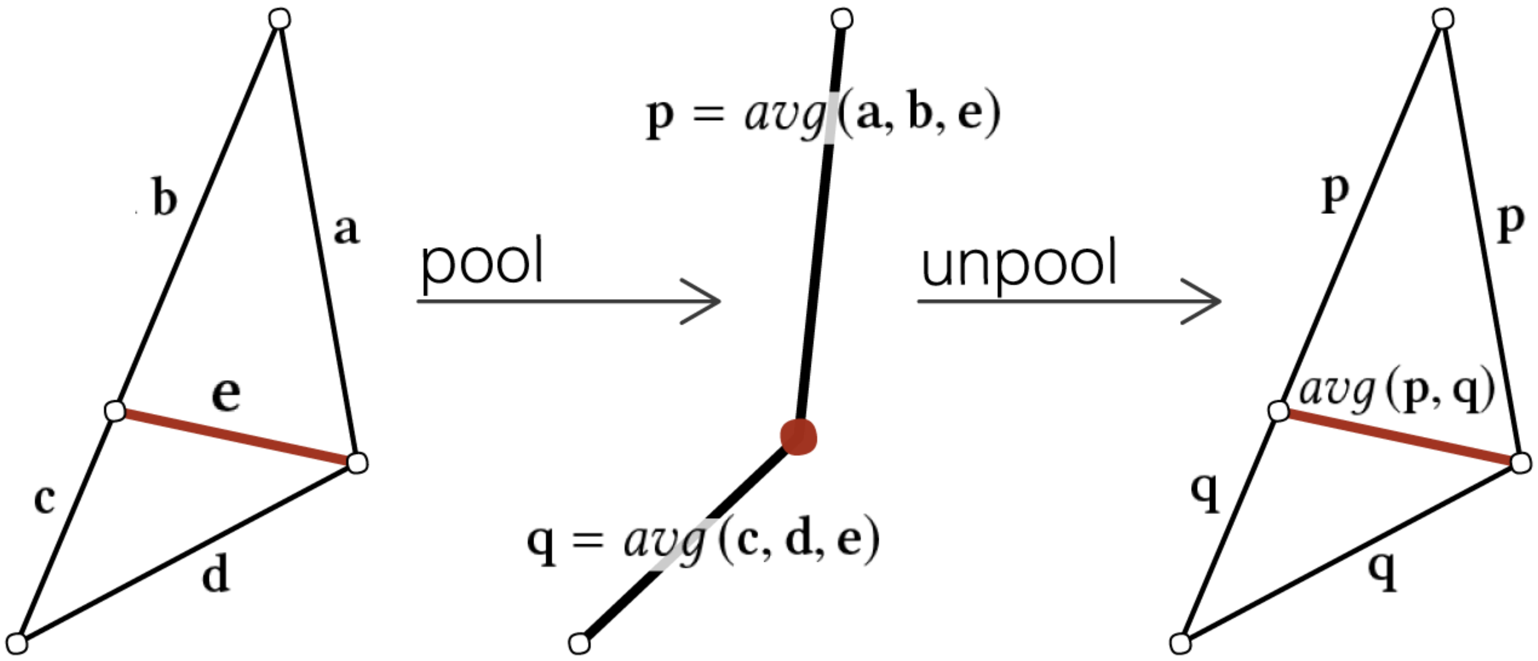
https://doc.cgal.org/latest/Surface_mesh_simplification/index.html



Yao et al., Quadratic Error Metric Mesh Simplification Algorithm Based on Discrete Curvature, 2015

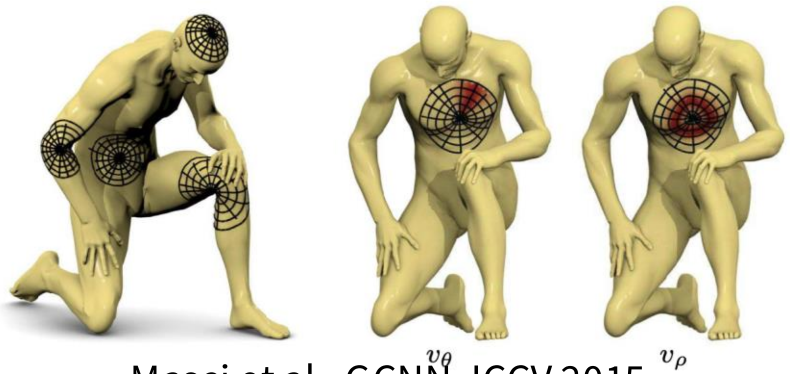
MeshCNN [Hanoka et al., 2019]

Pool adjacent edge information via the edge contraction.

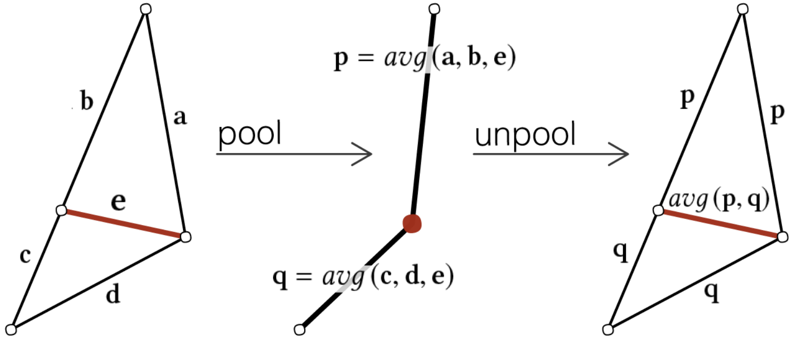


Neural Networks for Meshes

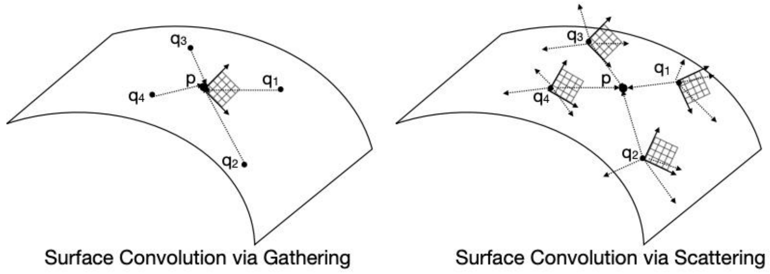
- Requires parameterization or specialized operations.
- Hard to implement. Verified only with a few use cases.



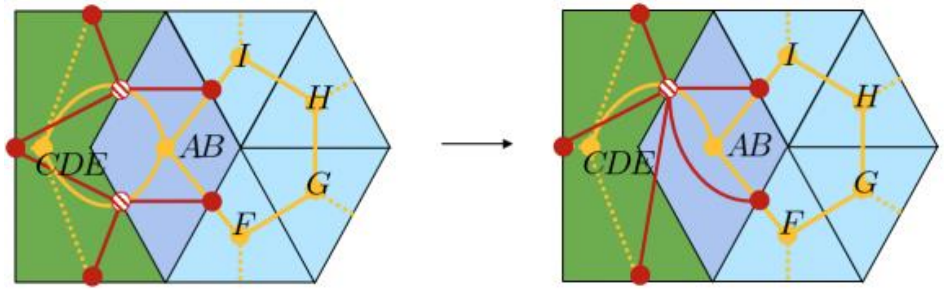
Masci et al., GCNN, ICCV 2015.



Hanocka et al., MeshCNN, SIGGRAPH 2019.



Mitchel et al., ICCV 2021.



Milano et al., NeurIPS 2020.

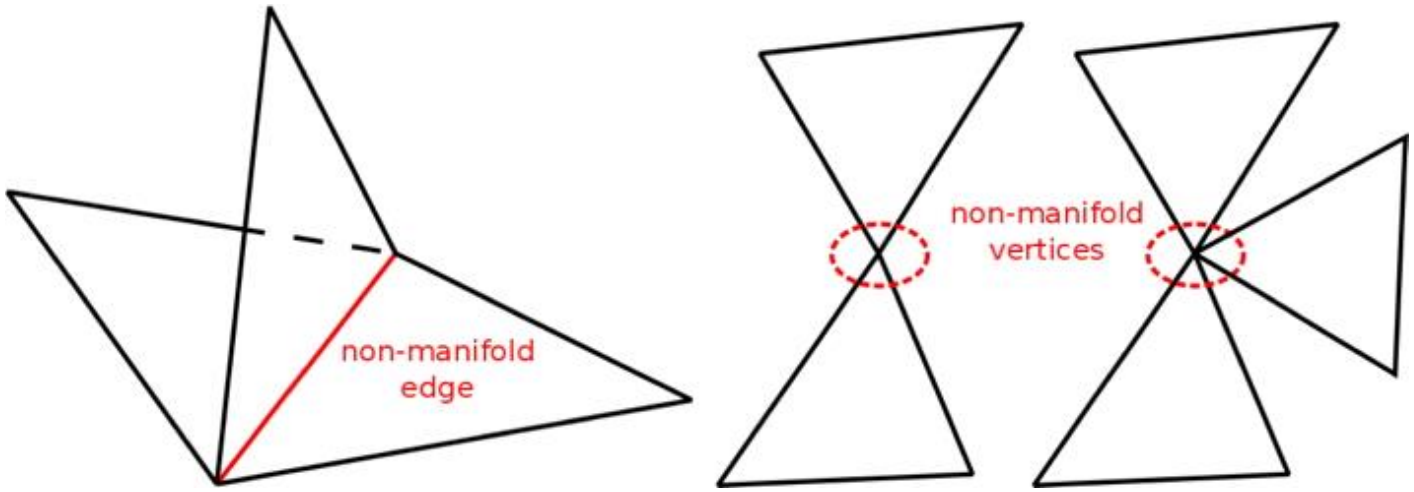
Polygon Mesh

- (+) Good for many applications:
  - Rendering
  - Texturing
  - Deformation / Manipulation
  - Simulation
- (–) Irregular structure
- (–) Difficult to create a valid mesh

Valid Meshes

E.g. 2-manifoldness

Each local region should be homeomorphic (mappable) to a 2D flat plane.

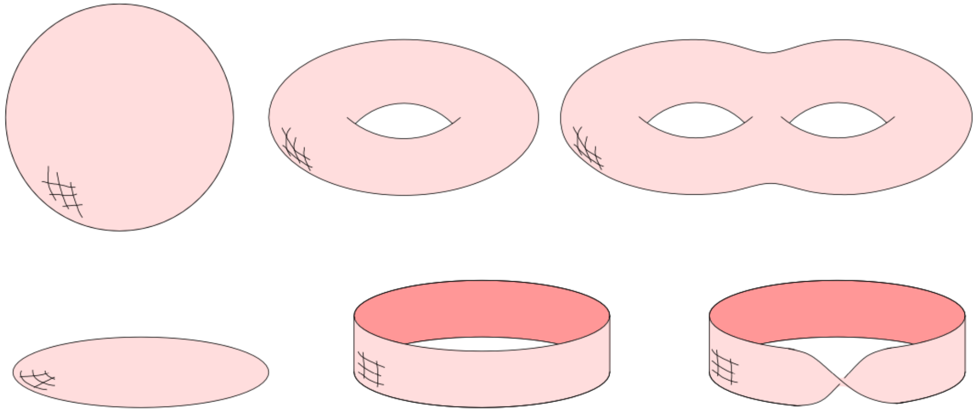


https://www.shapeways.com/blog/archives/29453-tutorial-tuesday-5-quick-fixes-with-meshlab.html

Valid Meshes

- Watertightness
- Topology
- Normal orientation consistency

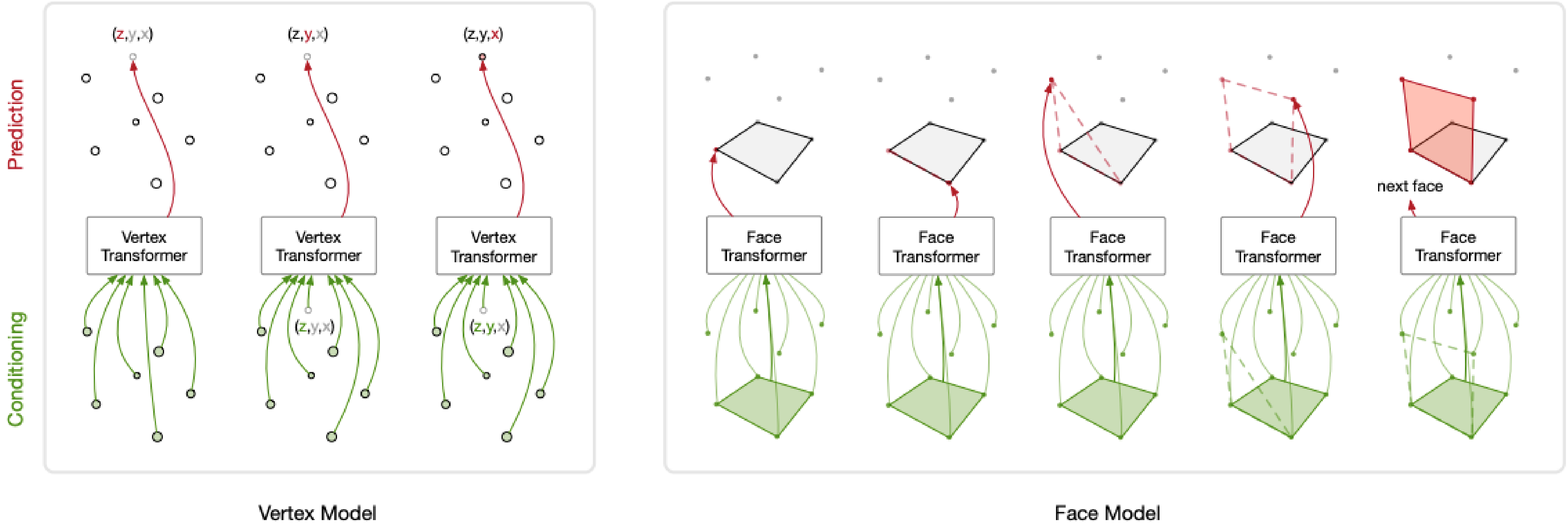
Etc.



https://courses.cs.duke.edu/fall06/cps296.1/Lectures/sec-II-1.pdf

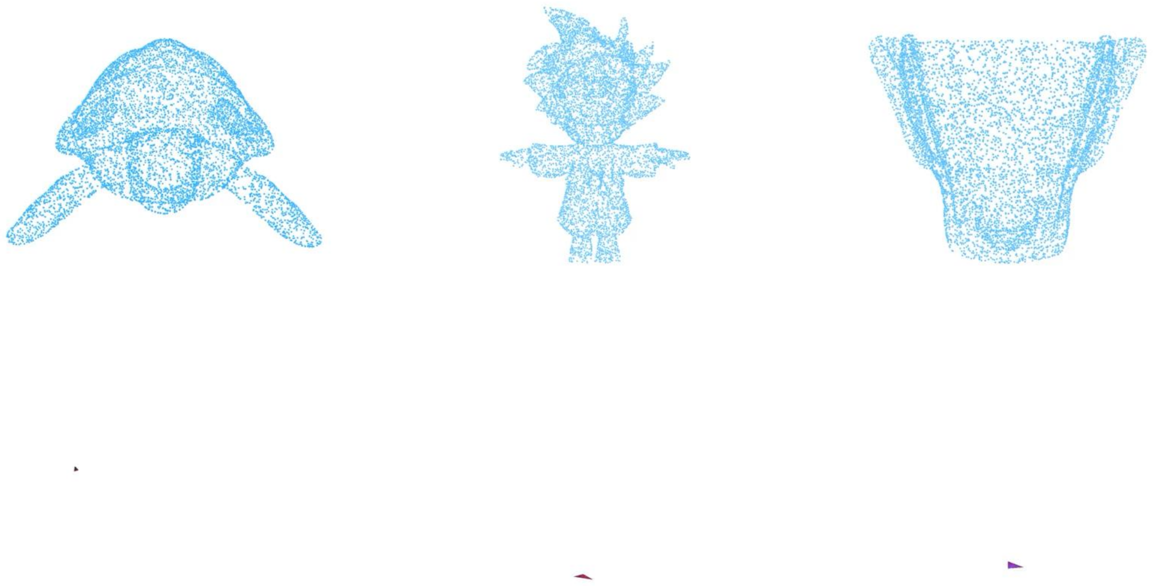
Mesh Generation

E.g., an autoregressive model generating vertices and faces sequentially.



Mesh Generation

E.g., an autoregressive model generating vertices and faces sequentially.

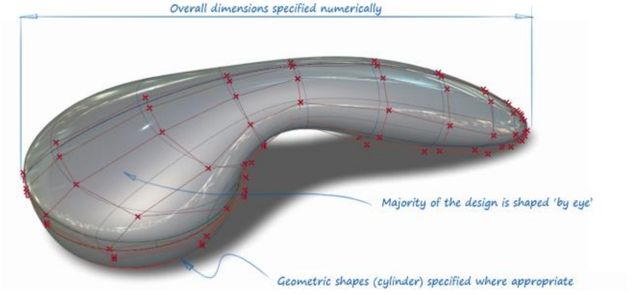


Point-Cloud Conditioned Mesh Generation Progress

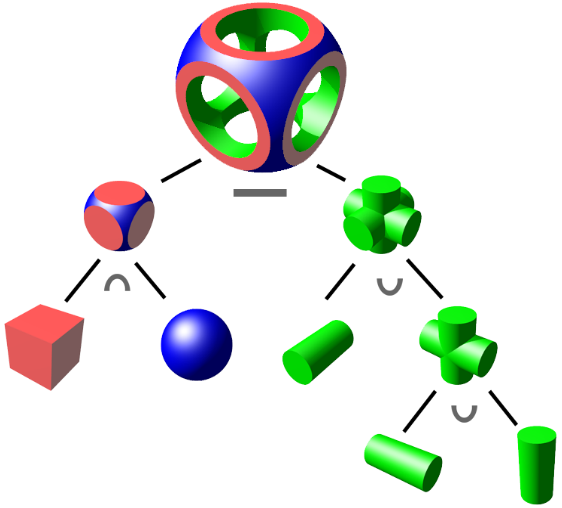
CAD Representations

CAD Representations

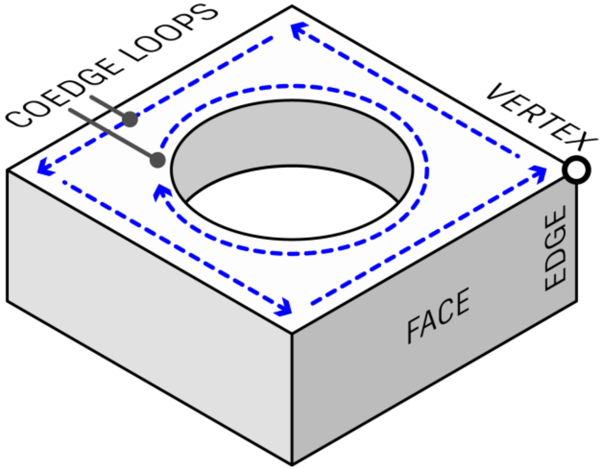
- More compact and strucural representations.
- NURBS, CSG, B-Rep, Extrusions, Revolve, etc.



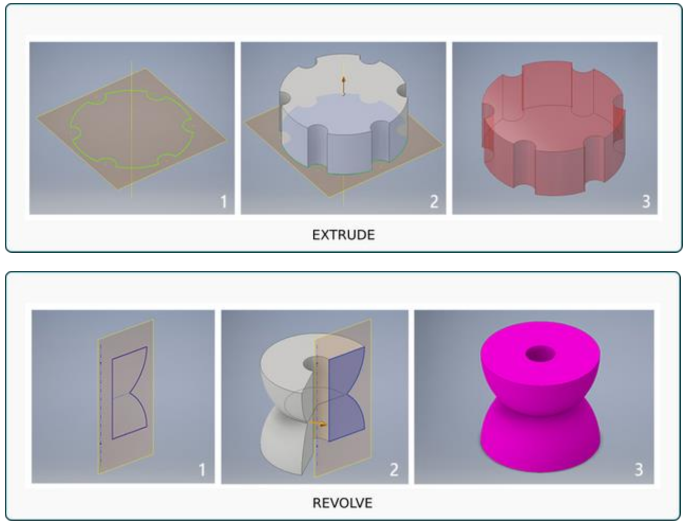
Autodesk



CSG



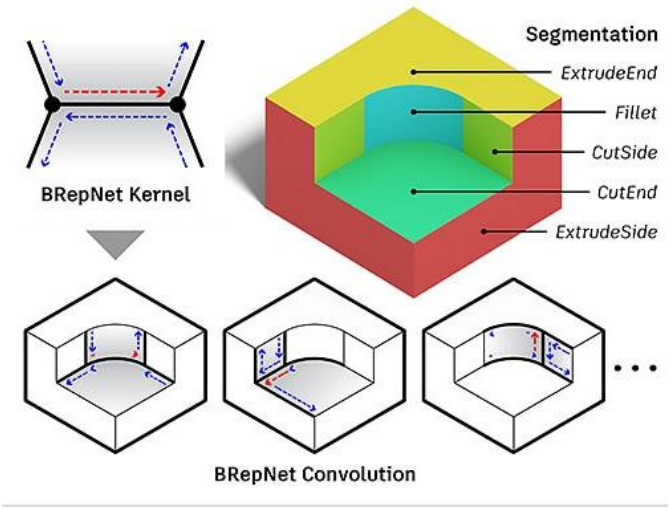
Karl D.D. Willis



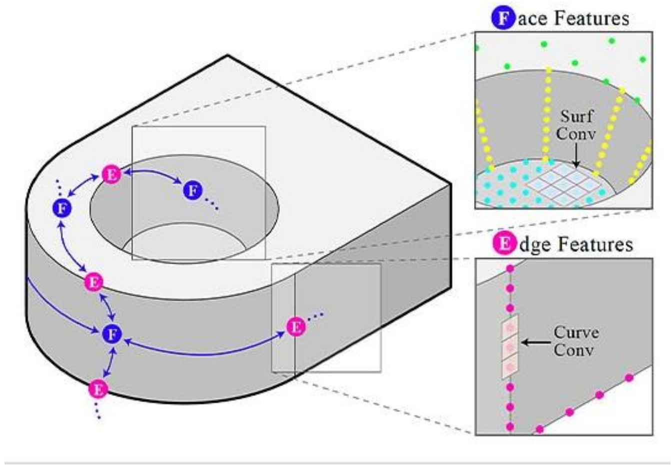
steemit

CAD Representations

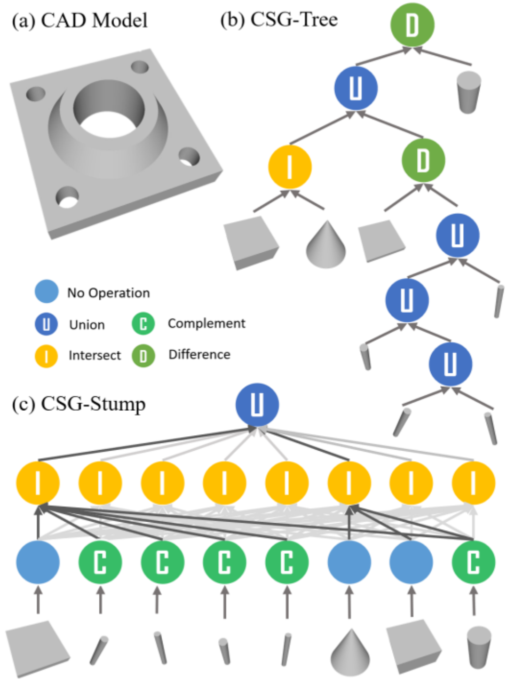
- Very few neural networks processing and generating them.



Lambourne et al., BRepNet, CVPR 2021.



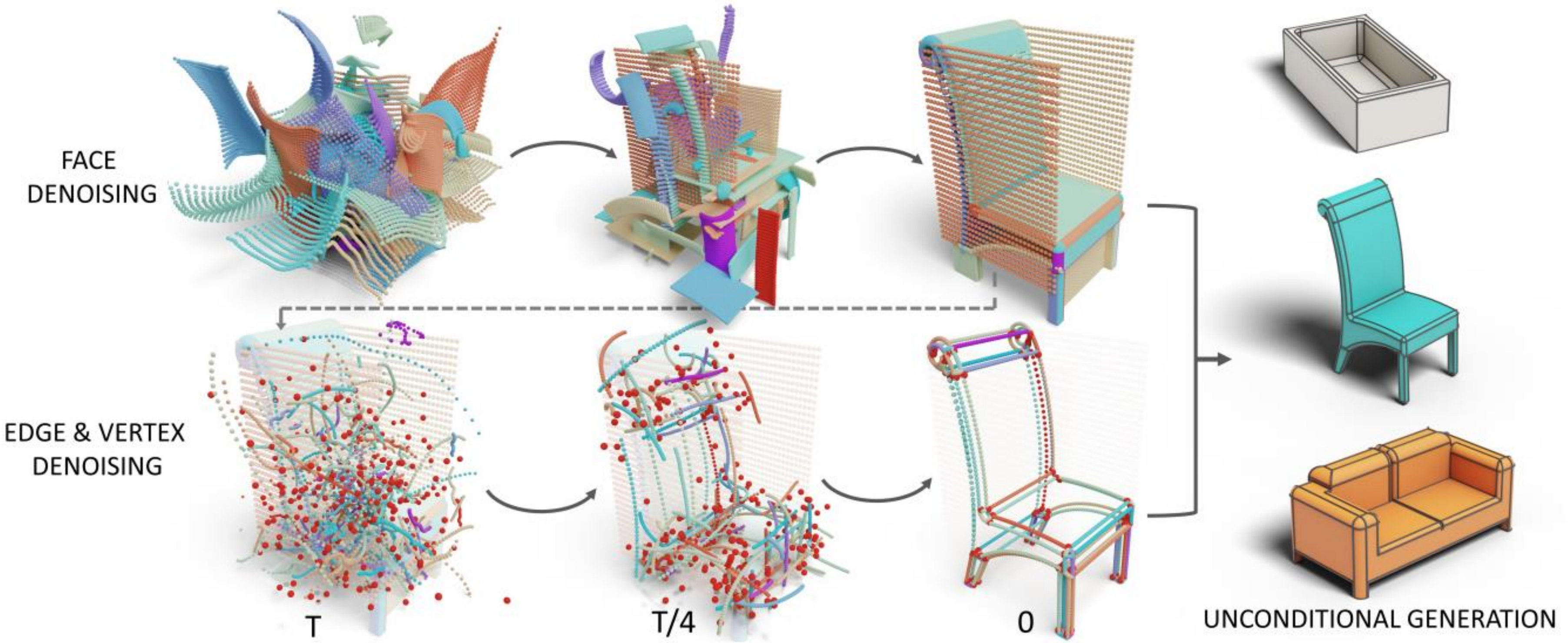
Jayaraman et al., UV-Net, CVPR 2021.



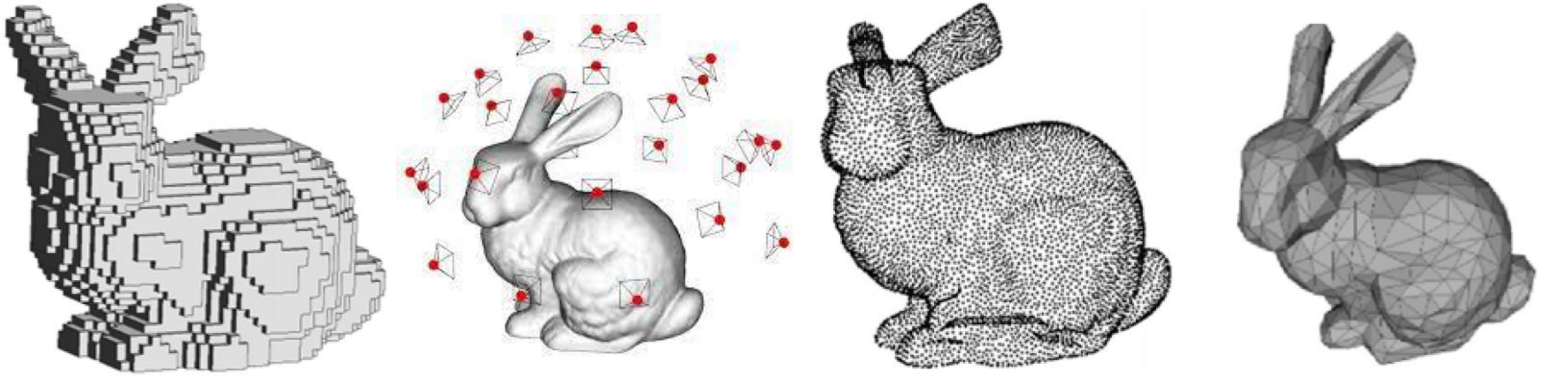
Ren et al., CSG-Stump, ICCV 2021.

CAD Representations

- Very few neural networks processing and generating them.
- The architecture are very complex.



3D Representations

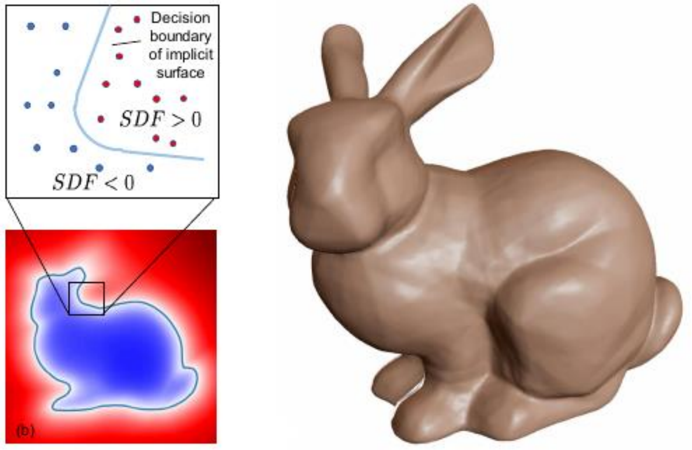


	Voxels	Multi-View	Point Cloud	Mesh
Pros	• Okay for processing if we use sparse convolution	• Good for processing • Easy to implement	• Good for both processing and generation • Efficient • Easy to implement	• Compact • Good for many applications
Cons	• Hard to implement • Inefficient	• Inefficient	• Require lots of points to capture details	• Bad for both processing and generation

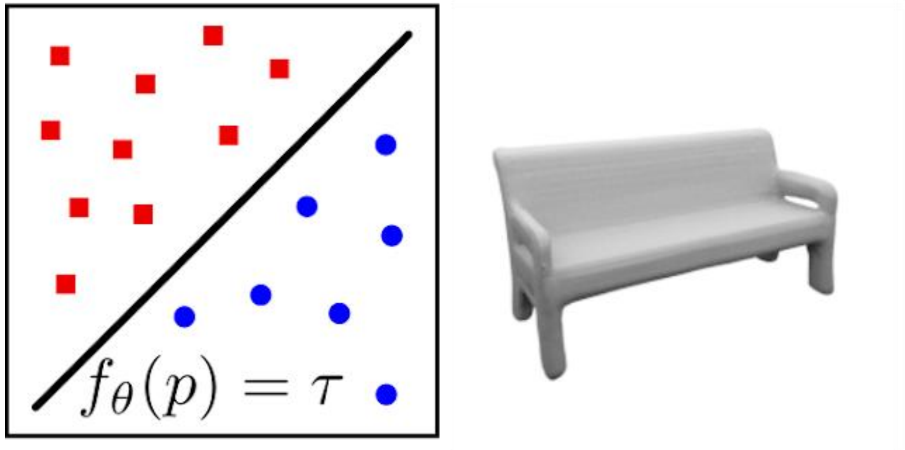
Implicit Representation

Implicit Representation

- A function that takes coordinates as input and returns occupancy or signed distance.
- Representation for output (generation) not input (processing).

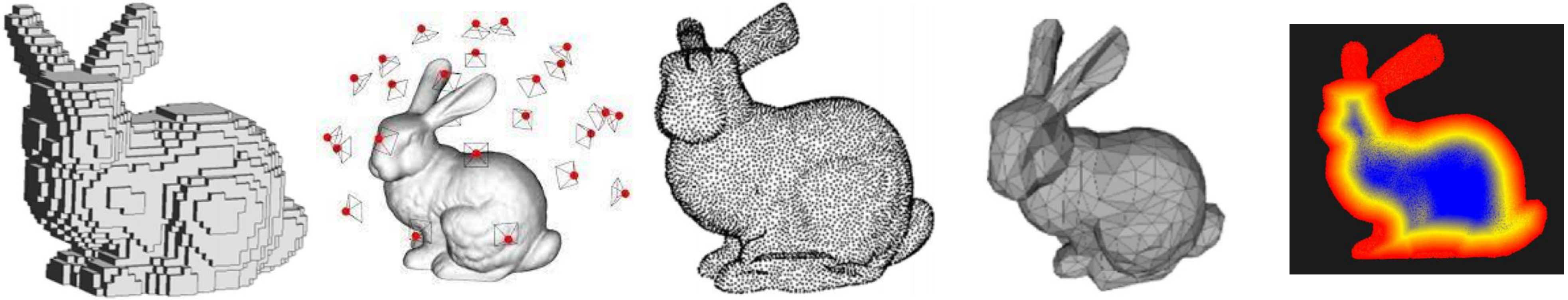


Park et al., DeepSDF, CVPR 2020.



Mescheder et al., Occupancy Networks, CVPR 2020.

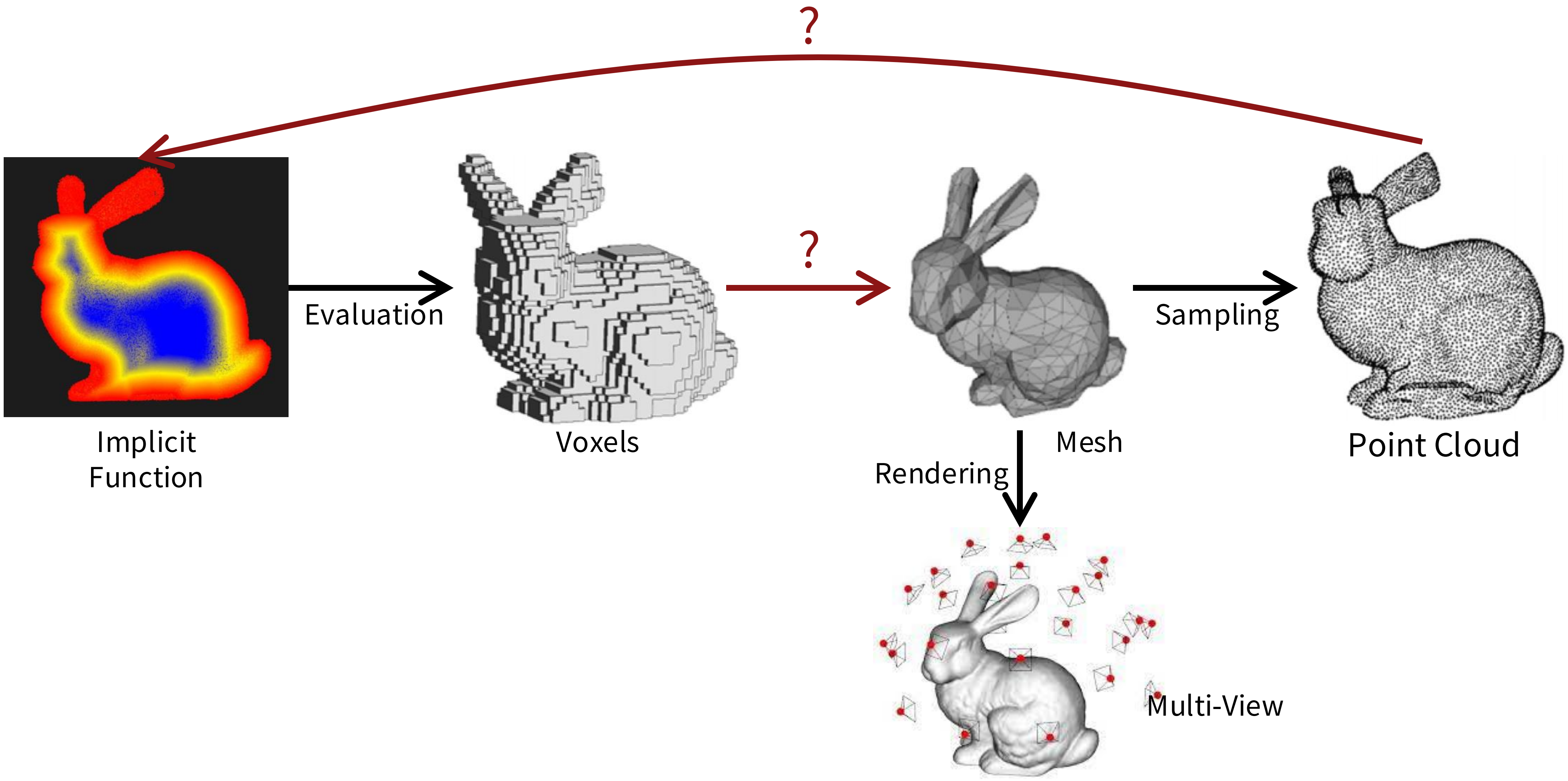
3D Representations



	Voxels	Multi-View	Point Cloud	Mesh	Implicit
Pros	• Okay for processing if we use sparse convolution	• Good for processing • Easy to implement	• Good for both processing and generation • Efficient • Easy to implement	• Compact • Good for many applications	• Good for generation
Cons	• Hard to implement • Inefficient	• Inefficient	• Require lots of points to capture details	• Bad for both processing and generation	• Cannot be used for processing • Need conversion for applications

Conversion Across Representations

Conversion Across Representations



Course Road Map

Representations

- Point clouds
- Implicit representation
- Multi-view images to 3D
- Hybrid representations
- Meshes
- ~~CAD~~
- Representation Conversion

Applications

- 3D perception (Encoding)
- Reconstruction (Decoding)
- Manipulation
- ~~Generation~~